
CSC-335

HCI & Computer

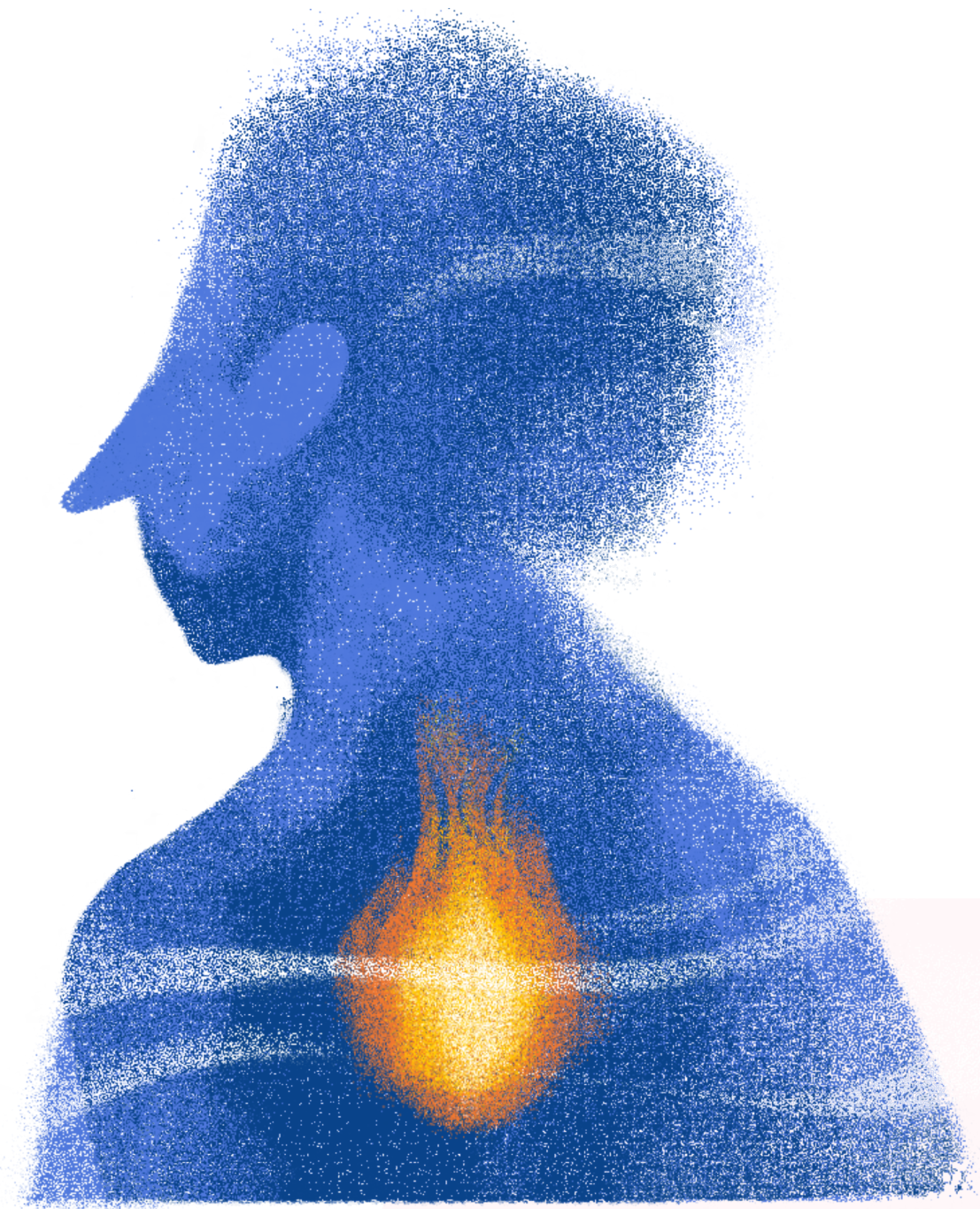
Graphics

Lec 2: Human Cognition in HCI



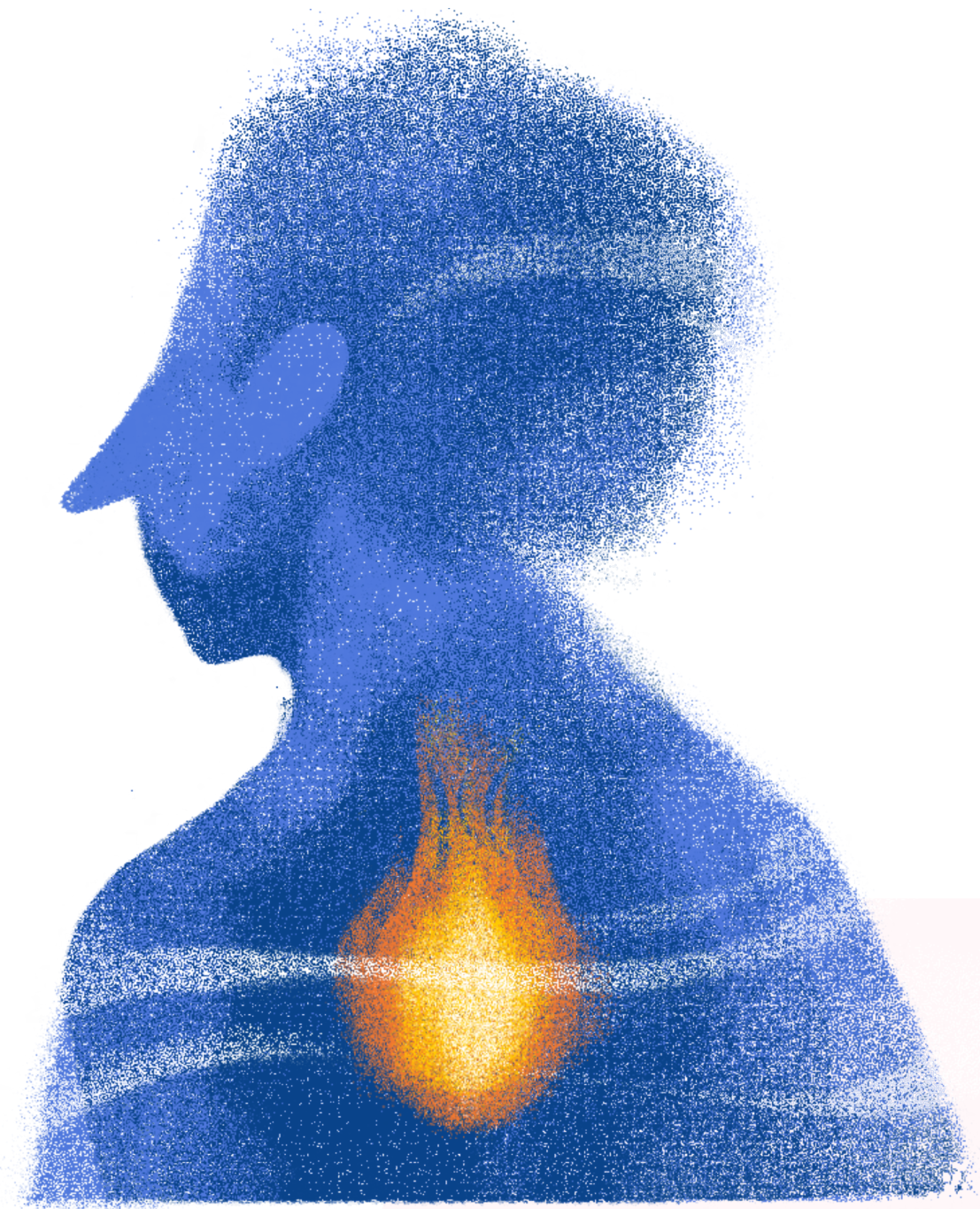
Introduction

- **Why do smart people still press the wrong button in apps?**
- **Why do users blame themselves instead of bad design?**



Introduction

- Why do smart people still press the wrong button in apps?
- Why do users blame themselves instead of bad design?



• Most user errors are design problems, not intelligence problems

Design must match how humans

Perceive

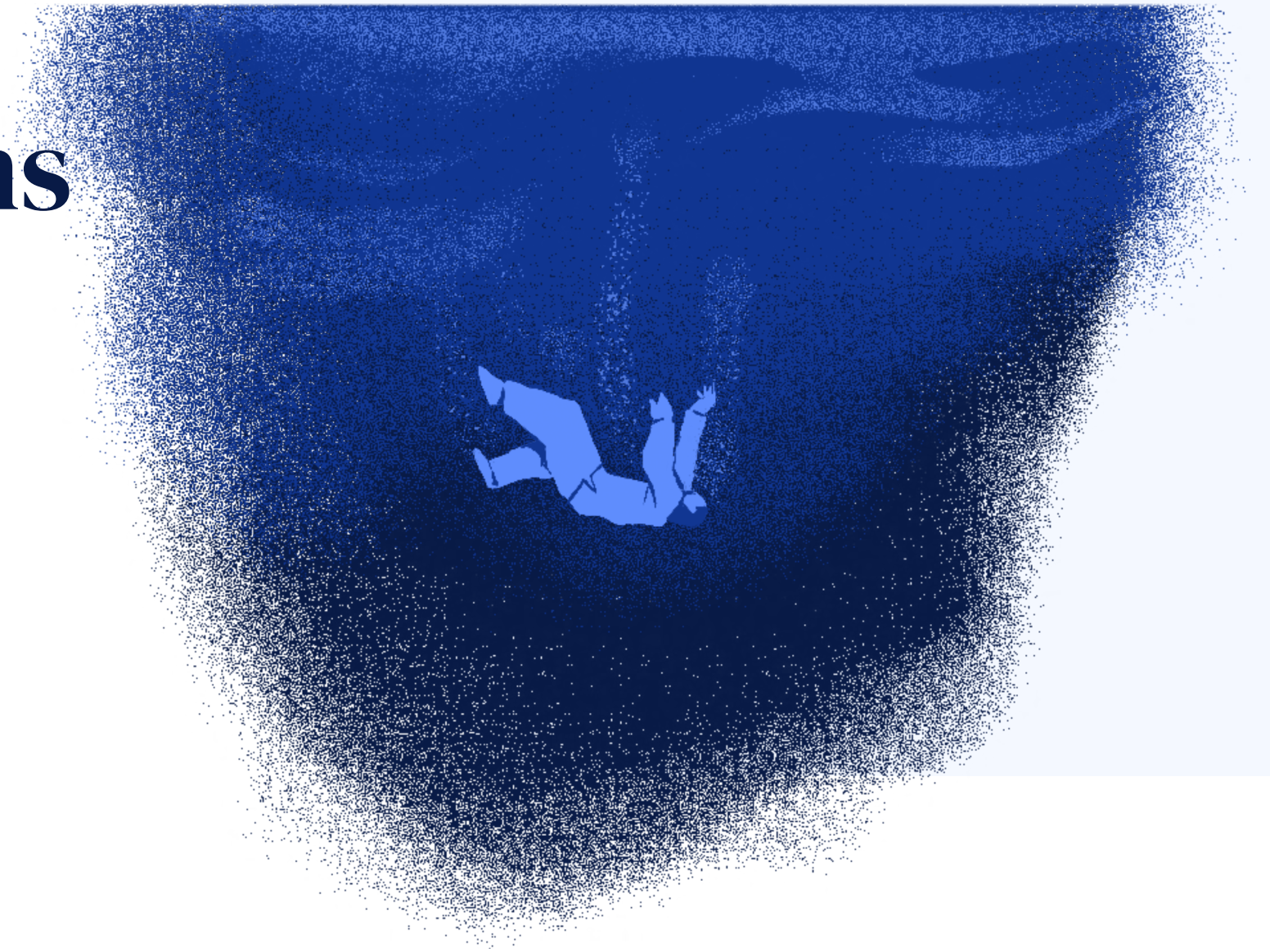
Attend

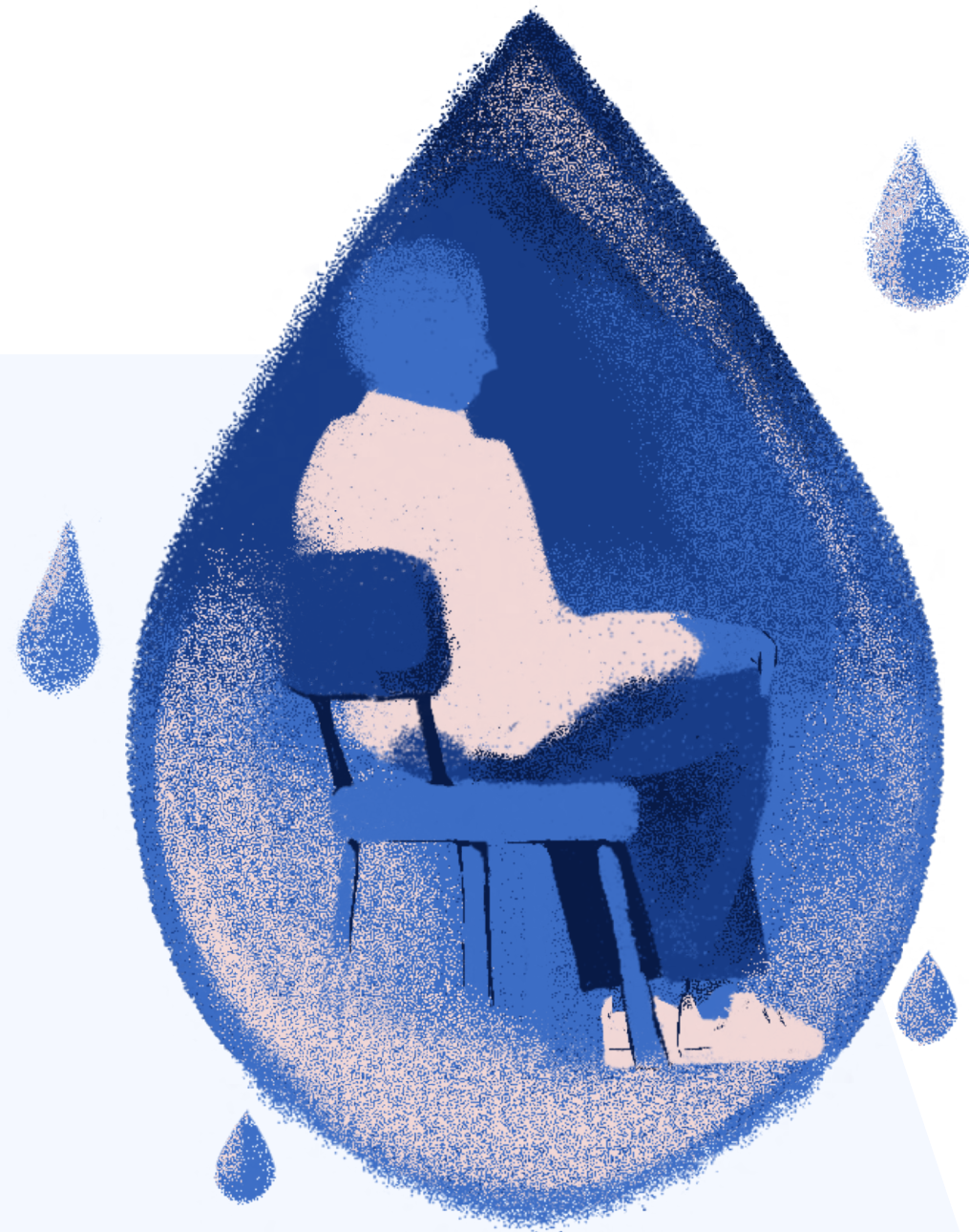
Remember

Decide

Act

Evaluate





01 Perception

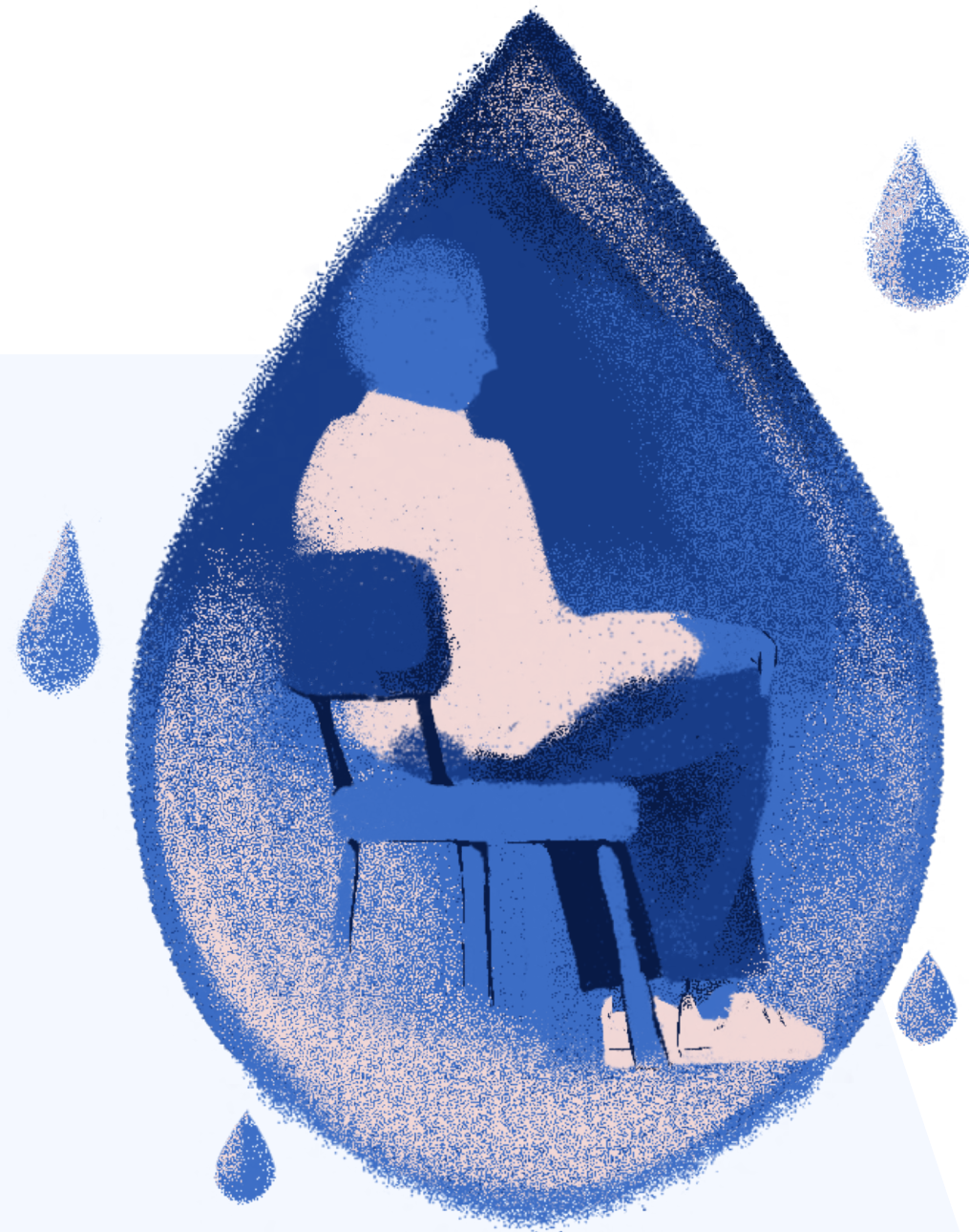
How humans interpret sensory input

NOT the same as seeing

Context comes into play

We don't see reality

We see *interpreted* reality



01 Perception

Key perception principles

Pattern Recognition

Gestalt Principles

Contrast & Visibility

Humans recognize patterns faster than reading text.

**Icons faster than labels
Logos faster than names**

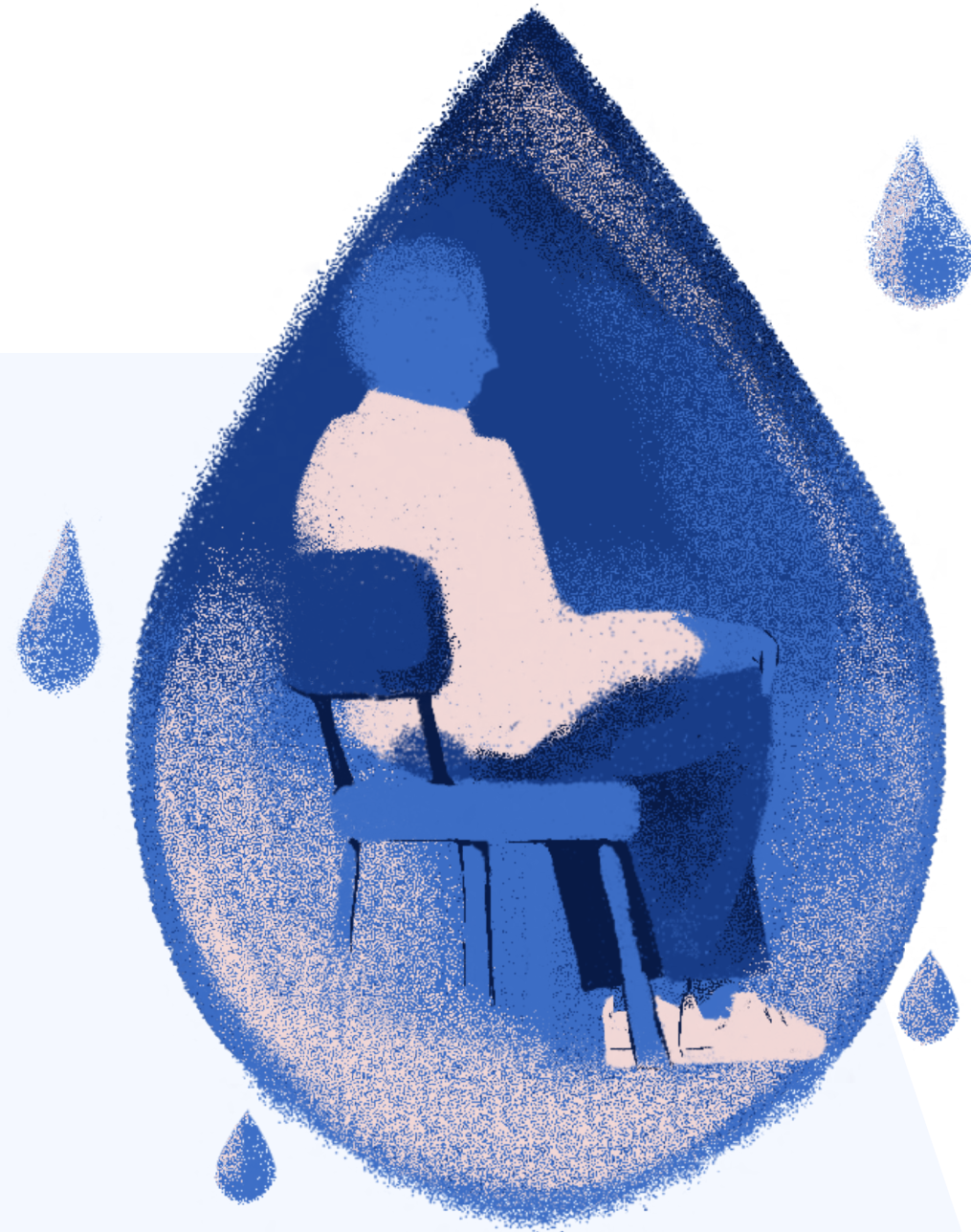
01 Perception

Key perception principles

Pattern Recognition

Gestalt Principles

Contrast & Visibility



Gray text on gray background vs High contrast
Tiny clickable links
Visual hierarchy
Clear affordances

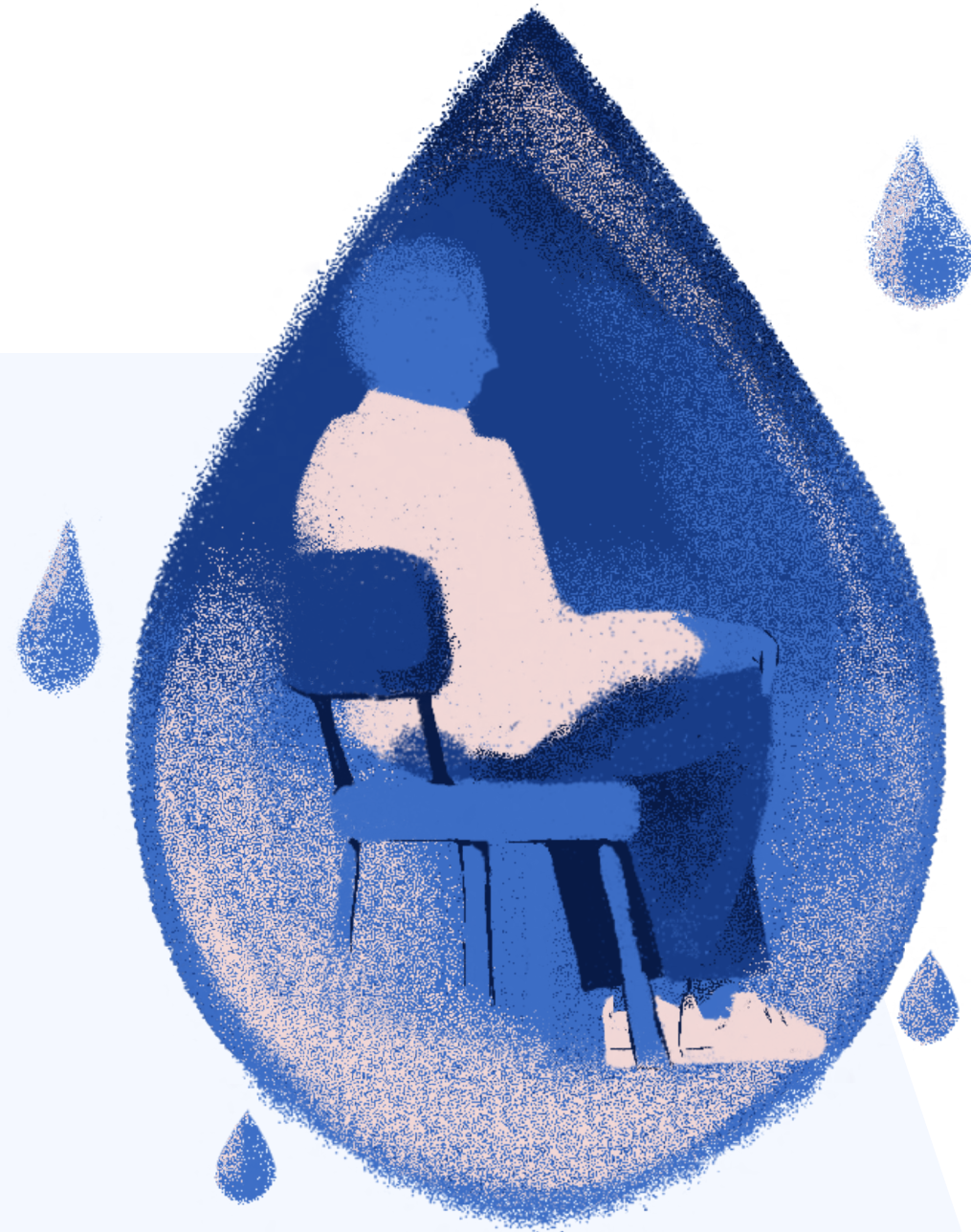
01 Perception

Key perception principles

Pattern Recognition

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Contrast & Visibility



Humans group visually by:
proximity, similarity, closure, continuity

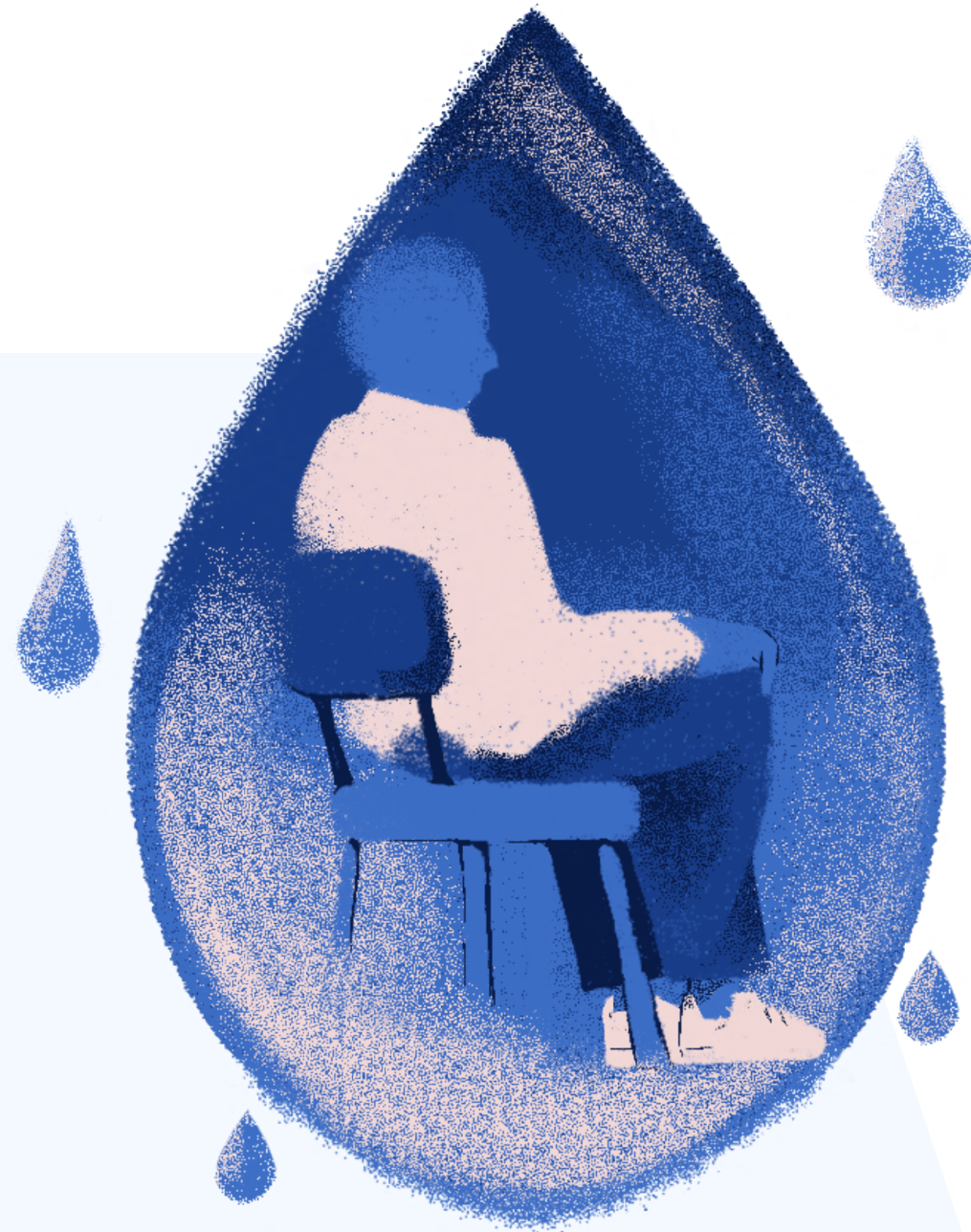
01 Perception

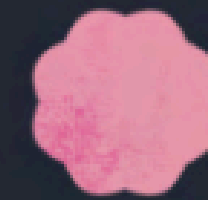
Key perception principles

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Gestalt Principles

Contrast & Visibility

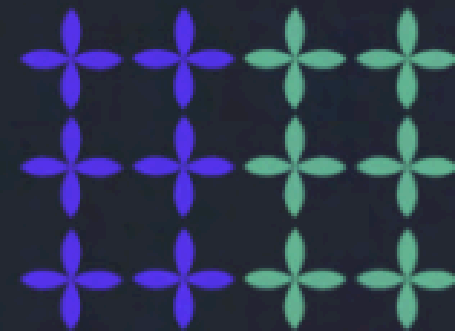




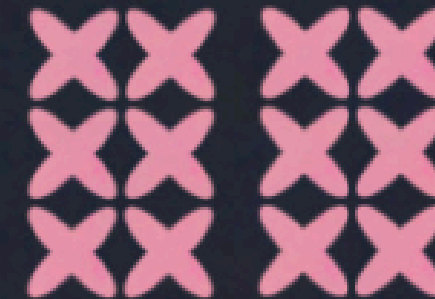
THE 6 PRINCIPLES OF GESTALT THEORY

shutterstock

1. SIMILARITY



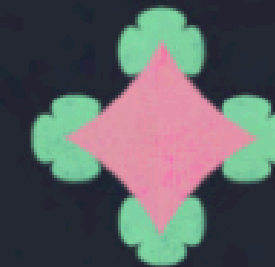
4. PROXIMITY



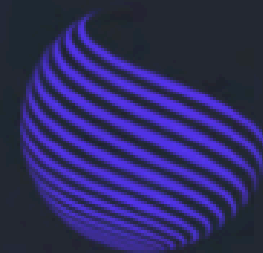
2. CONTINUATION



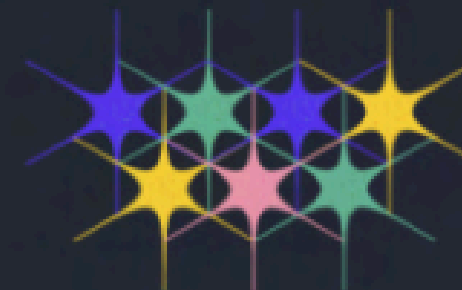
5. FIGURE/GROUND

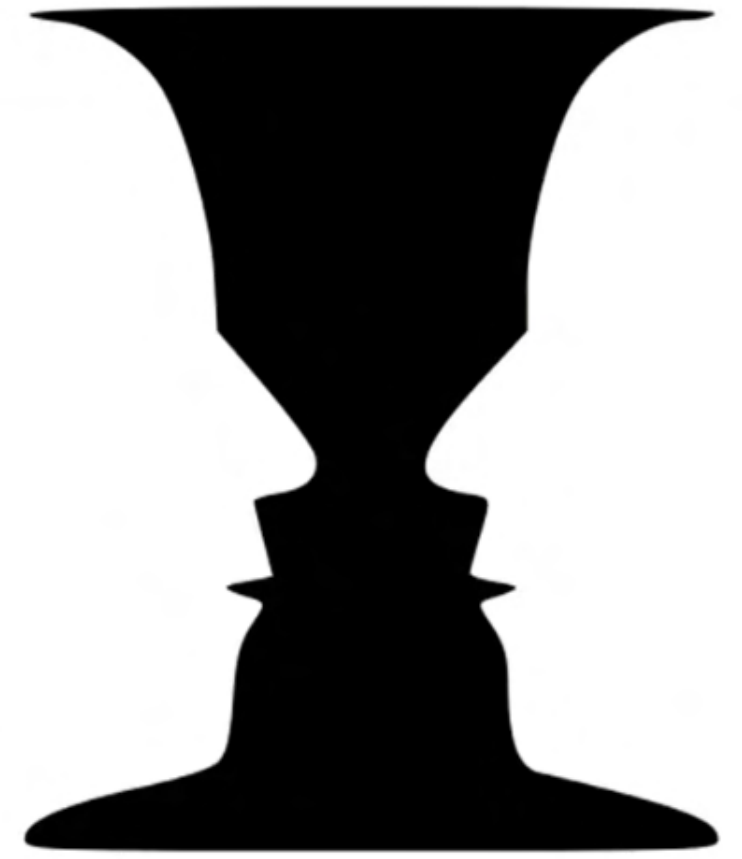


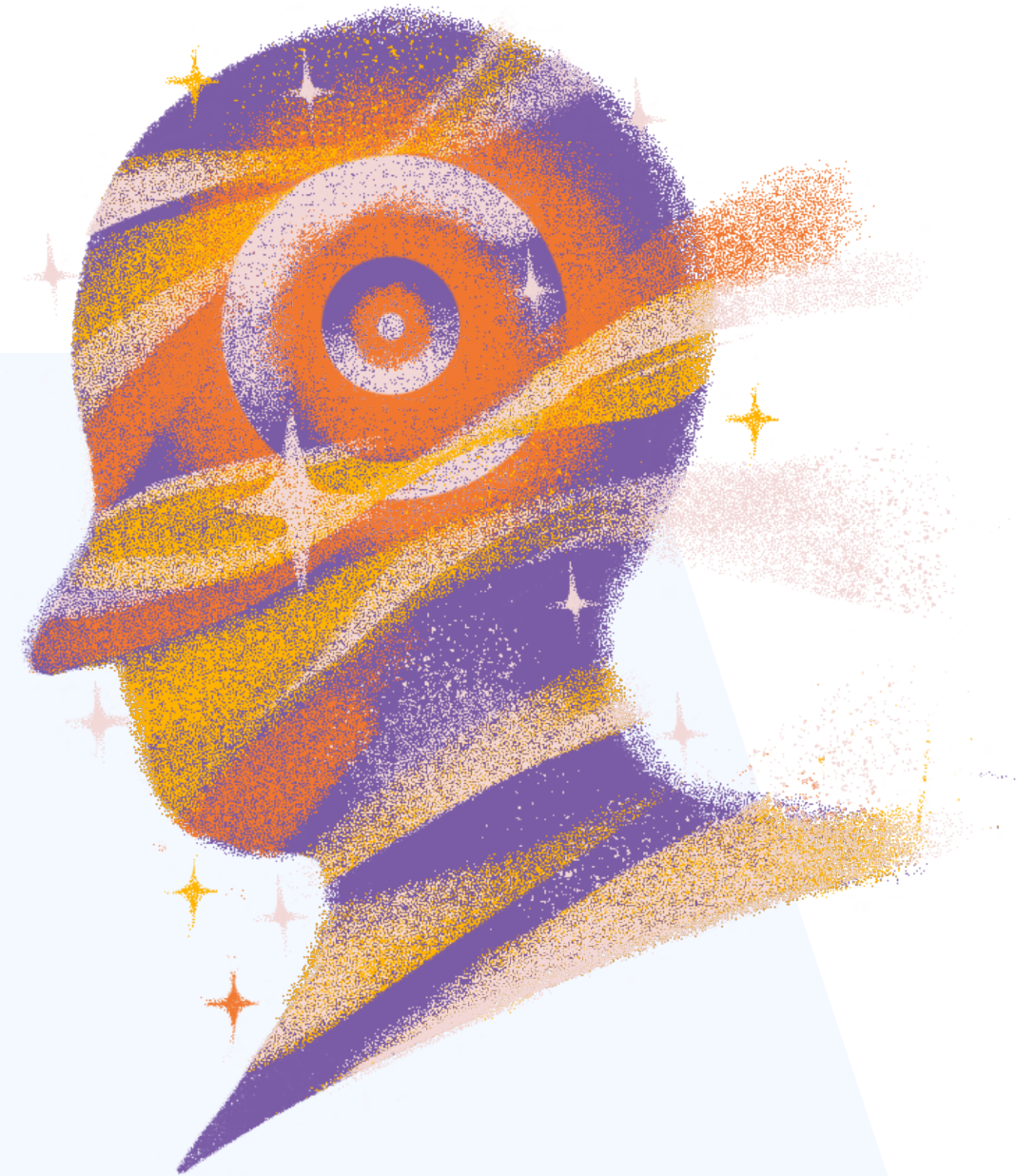
3. CLOSURE



6. PRÄGNANZ







02 Memory

**Short Term/Working
Memory**

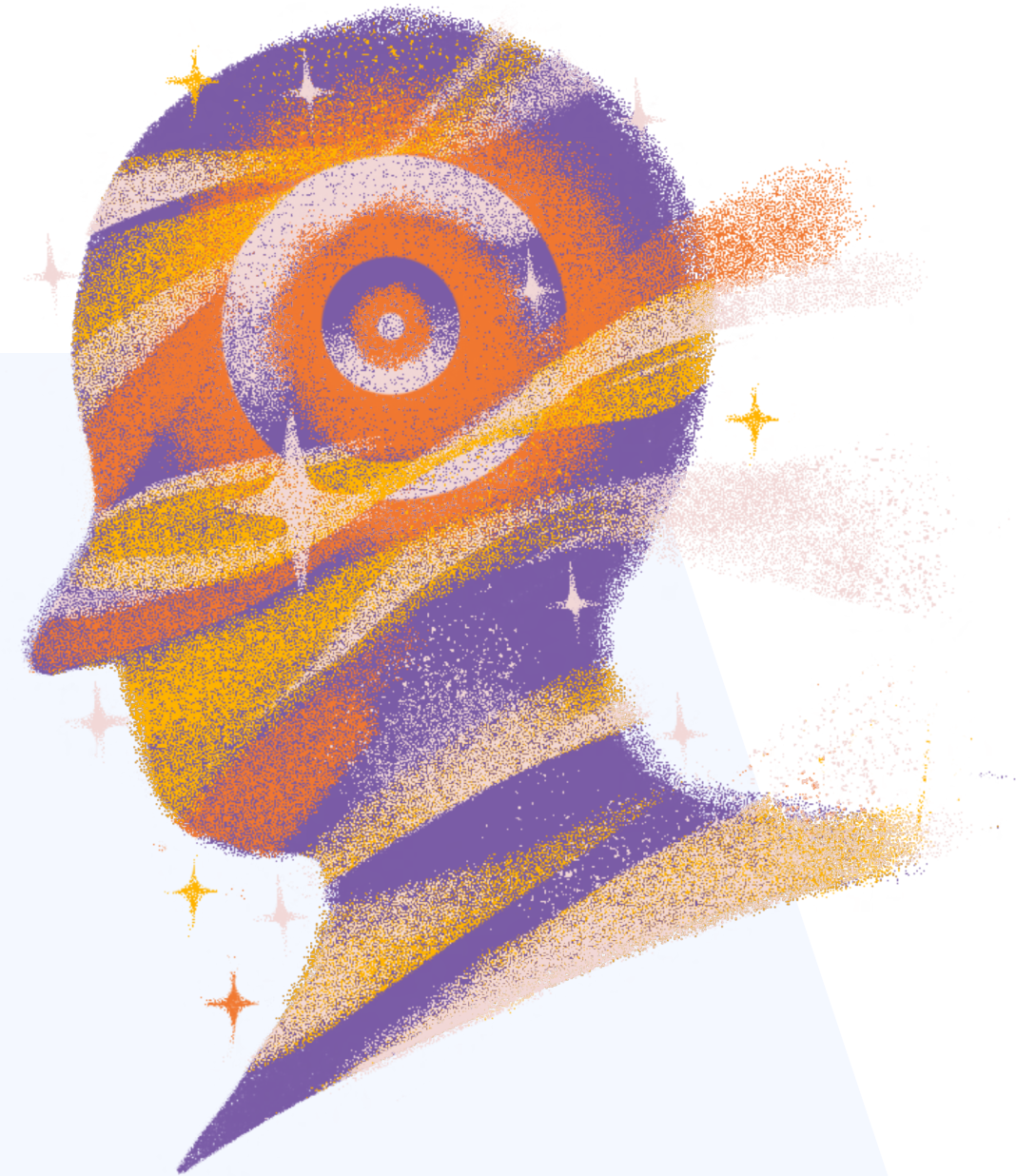
Recognition

Long-Term Memory

Recall

Capacity $\approx$ 4–7 items
Duration: seconds
Easily overloaded

02 Memory



**Short Term/Working
Memory**

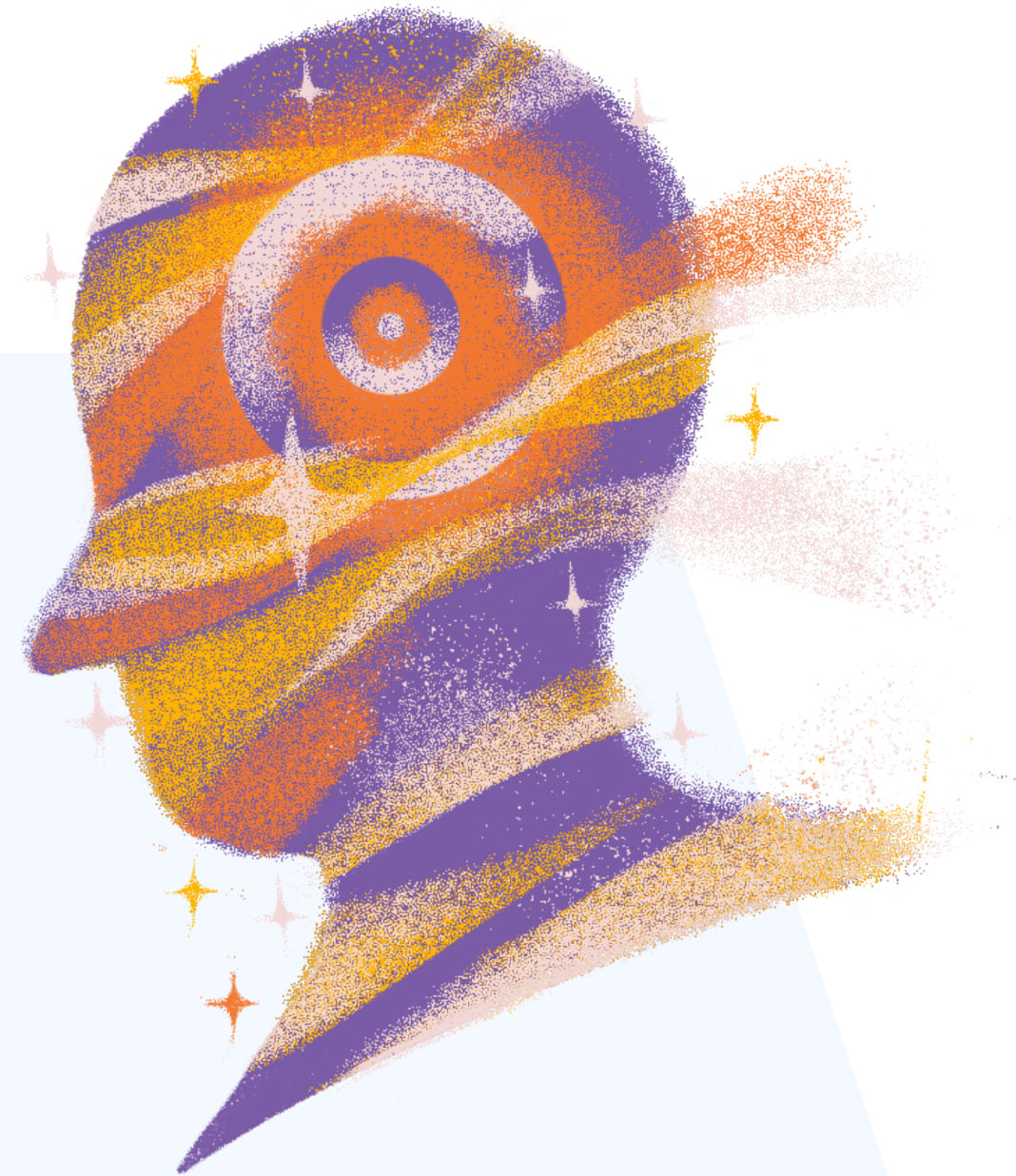
Recognition

Long-Term Memory

Recall

**Knowledge
Skills
Habits**

02 Memory



**Short Term/Working
Memory**

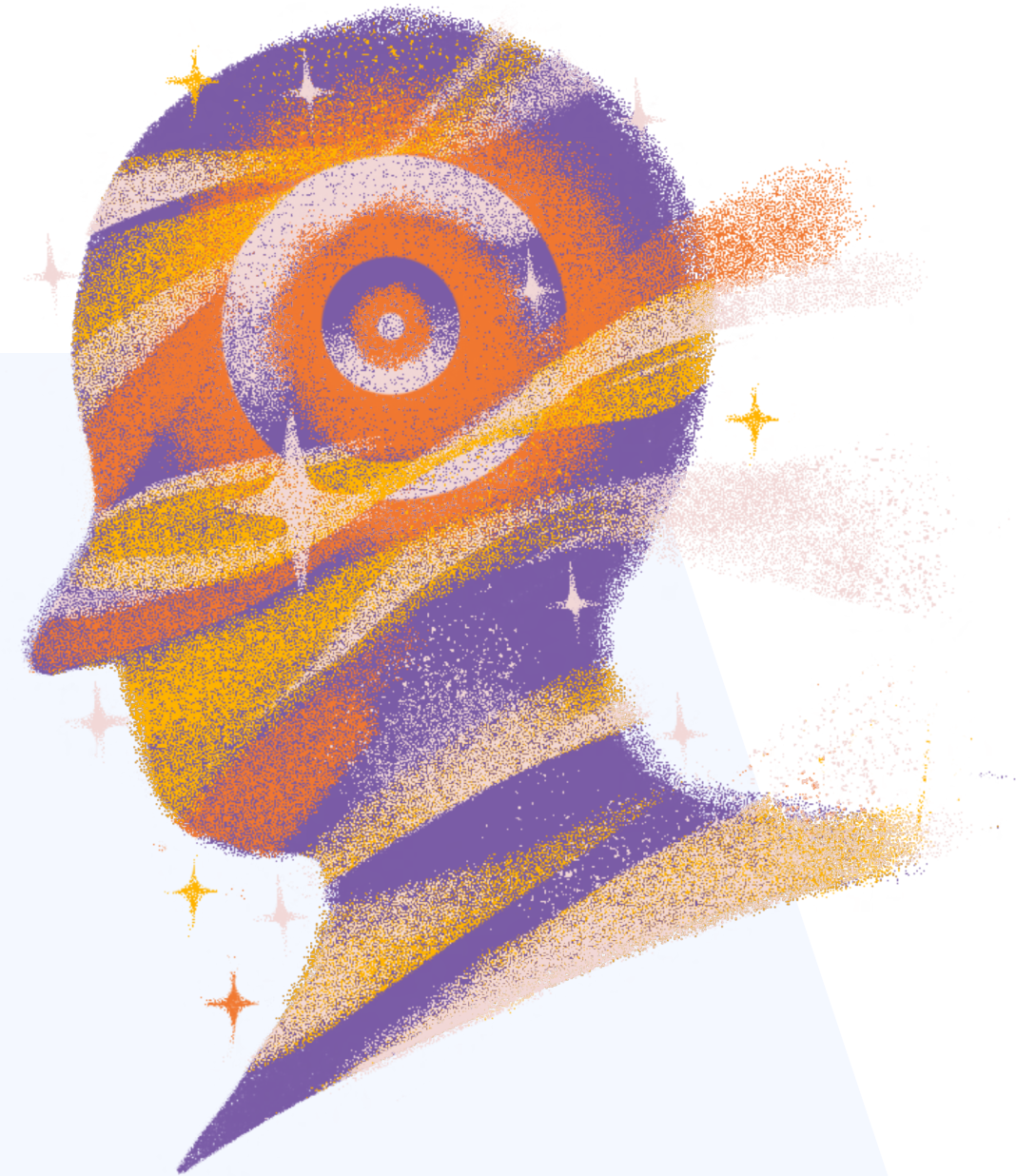
Long-Term Memory

Recognition

Recall

Choose from menu
Select icon
Pick from list

02 Memory



**Short Term/Working
Memory**

Recognition

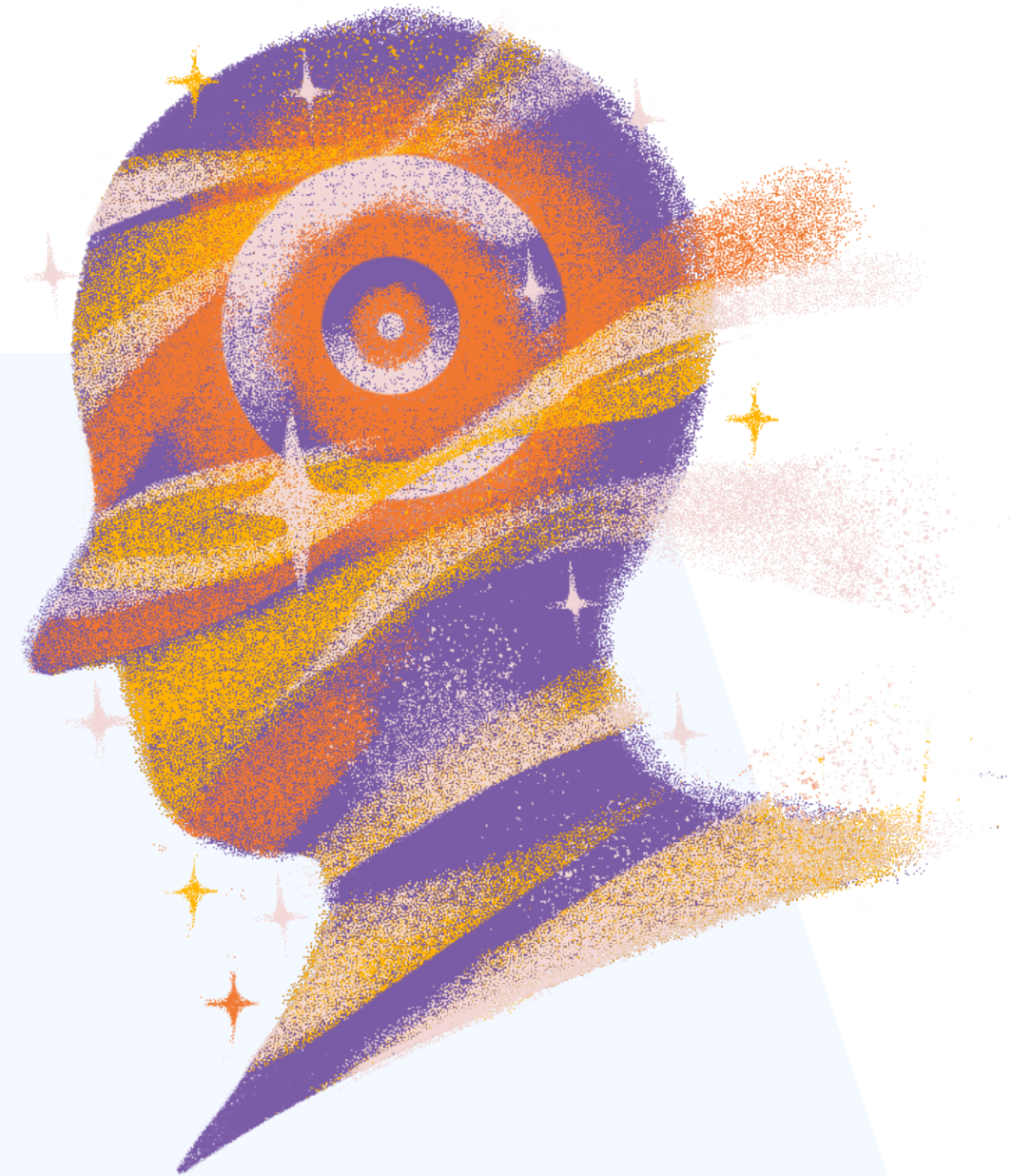
Long-Term Memory

Recall

Retrieving information from memory without
external cues

Type command, Remember syntax, Memorize
steps

02 Memory



Short Term/Working
Memory

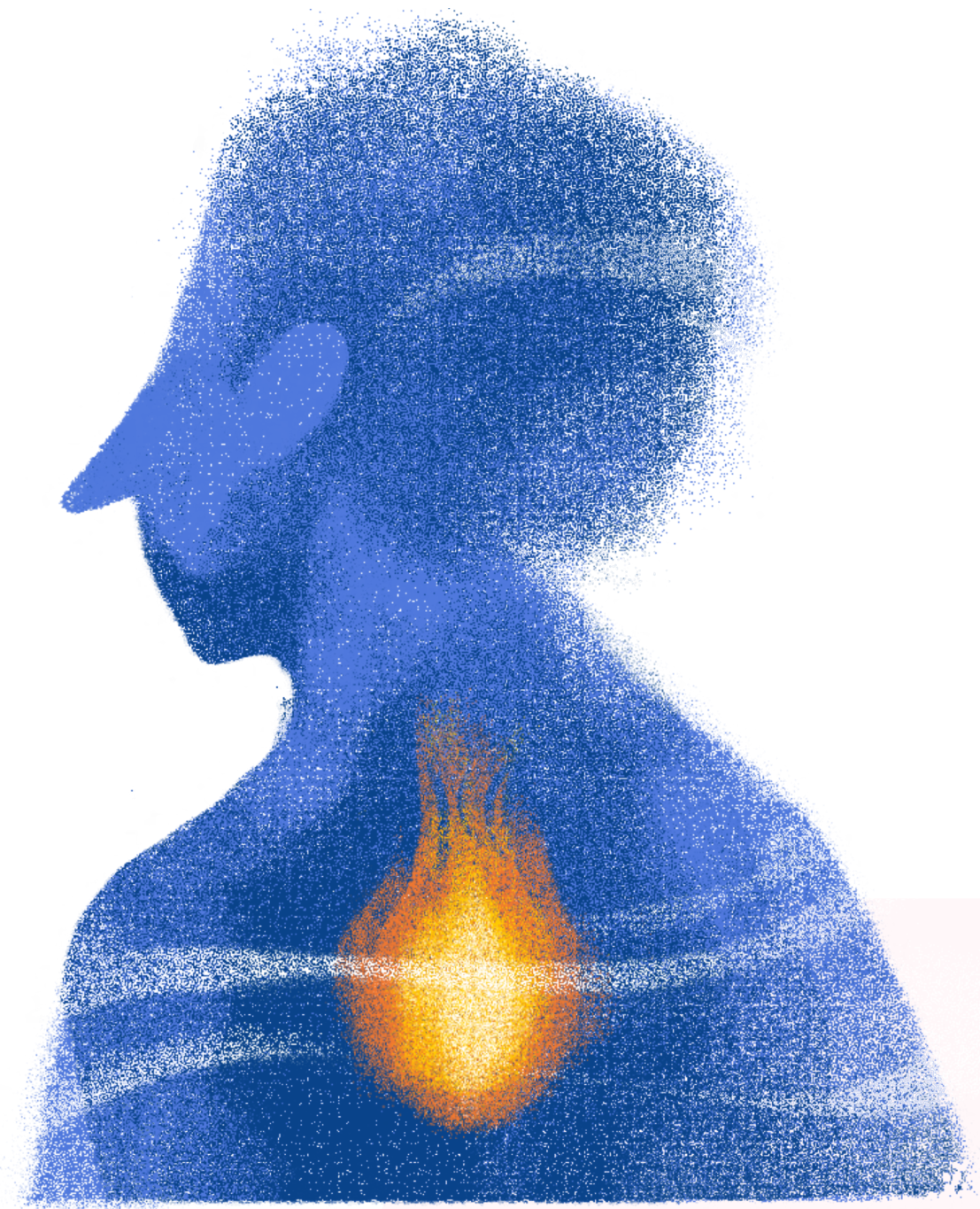
Long-Term Memory

Recognition

Recall

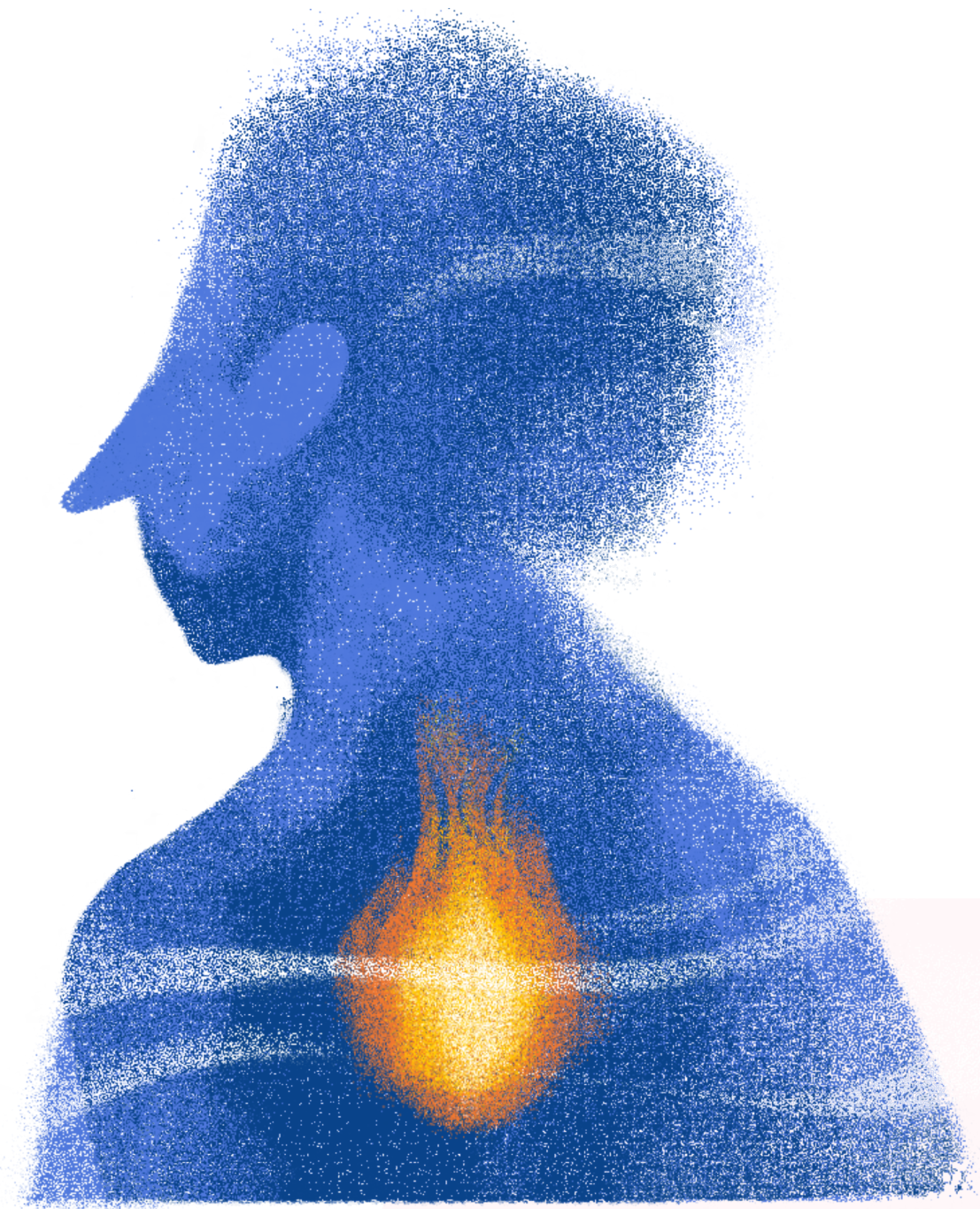
Discussion

**Why are command-line interfaces
harder for beginners?**



Discussion

Why are command-line interfaces harder for beginners?

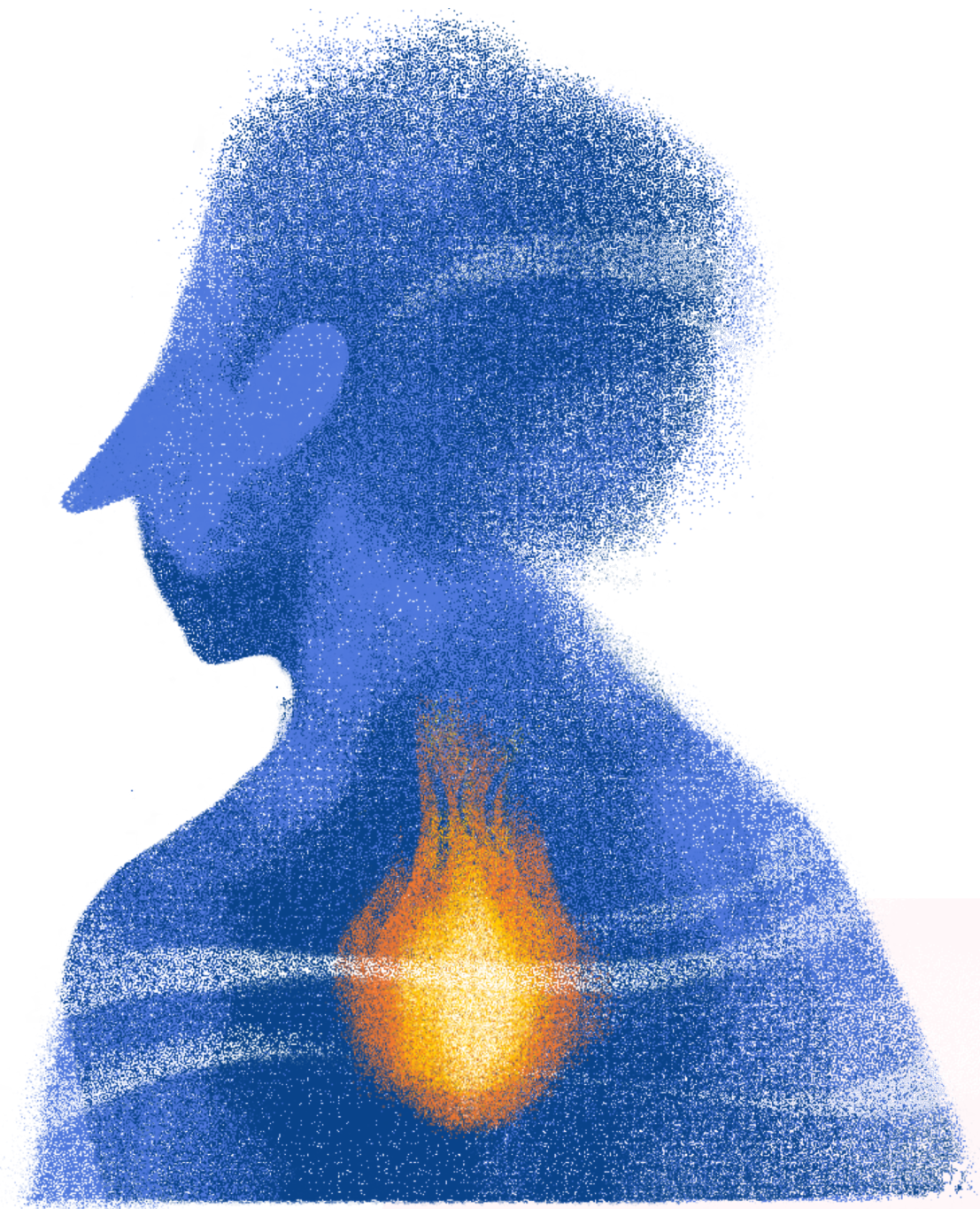


Recall burden, Syntax memory, No recognition support

Discussion

Which one of the better option?

Enter error code manually
Select error type from list





03 Attention

limited cognitive focus

Humans cannot process everything — they filter

Selective Attention

Divided Attention

Focus on one thing at a time
Reading text
ignoring background ads

03 Attention

limited cognitive focus

Humans cannot process everything — they filter

Selective Attention

Divided Attention



Multitasking → performance drops
Driving + texting
Coding + chatting

03 Attention

limited cognitive focus

Humans cannot process everything — they filter

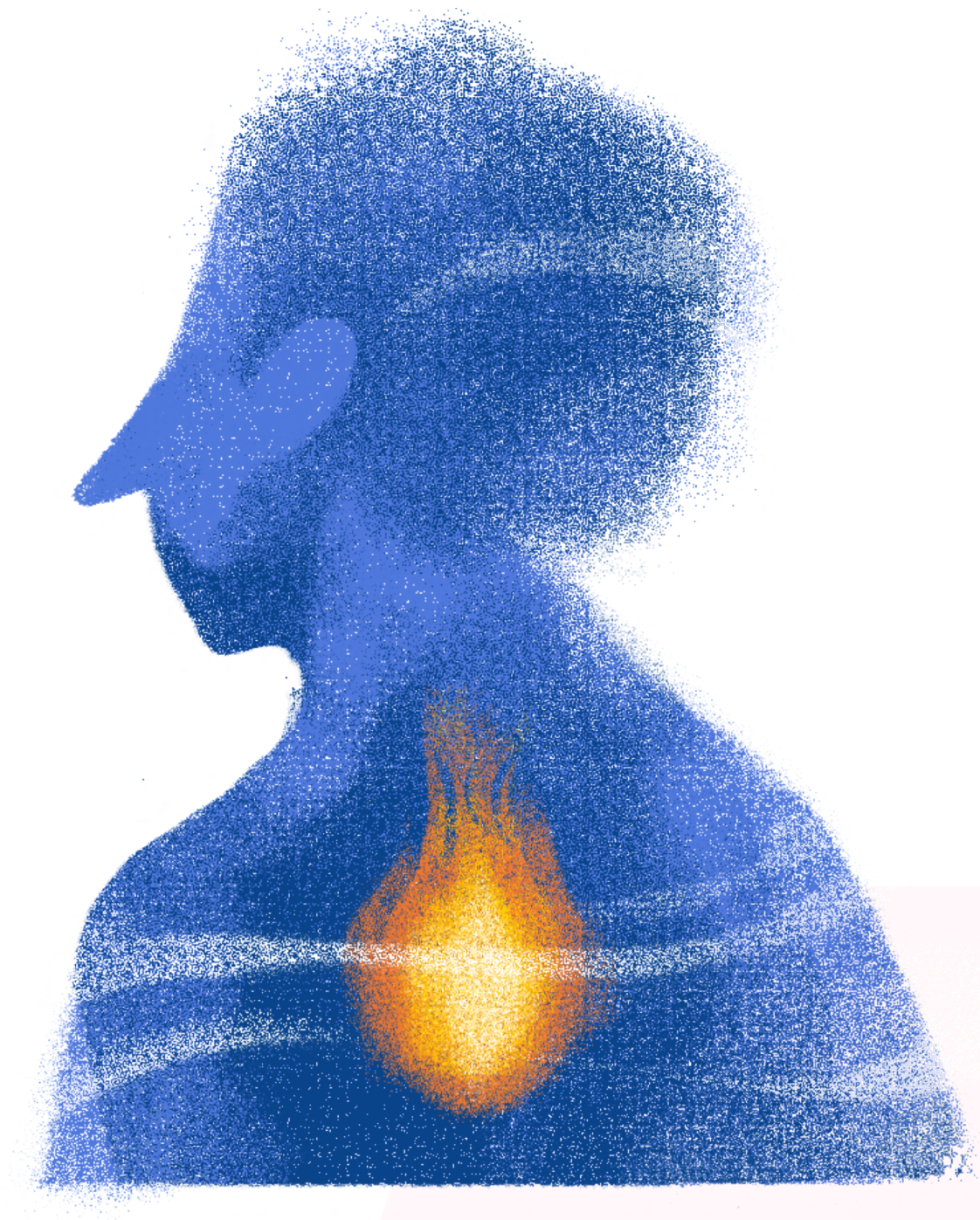
Selective Attention

Divided Attention



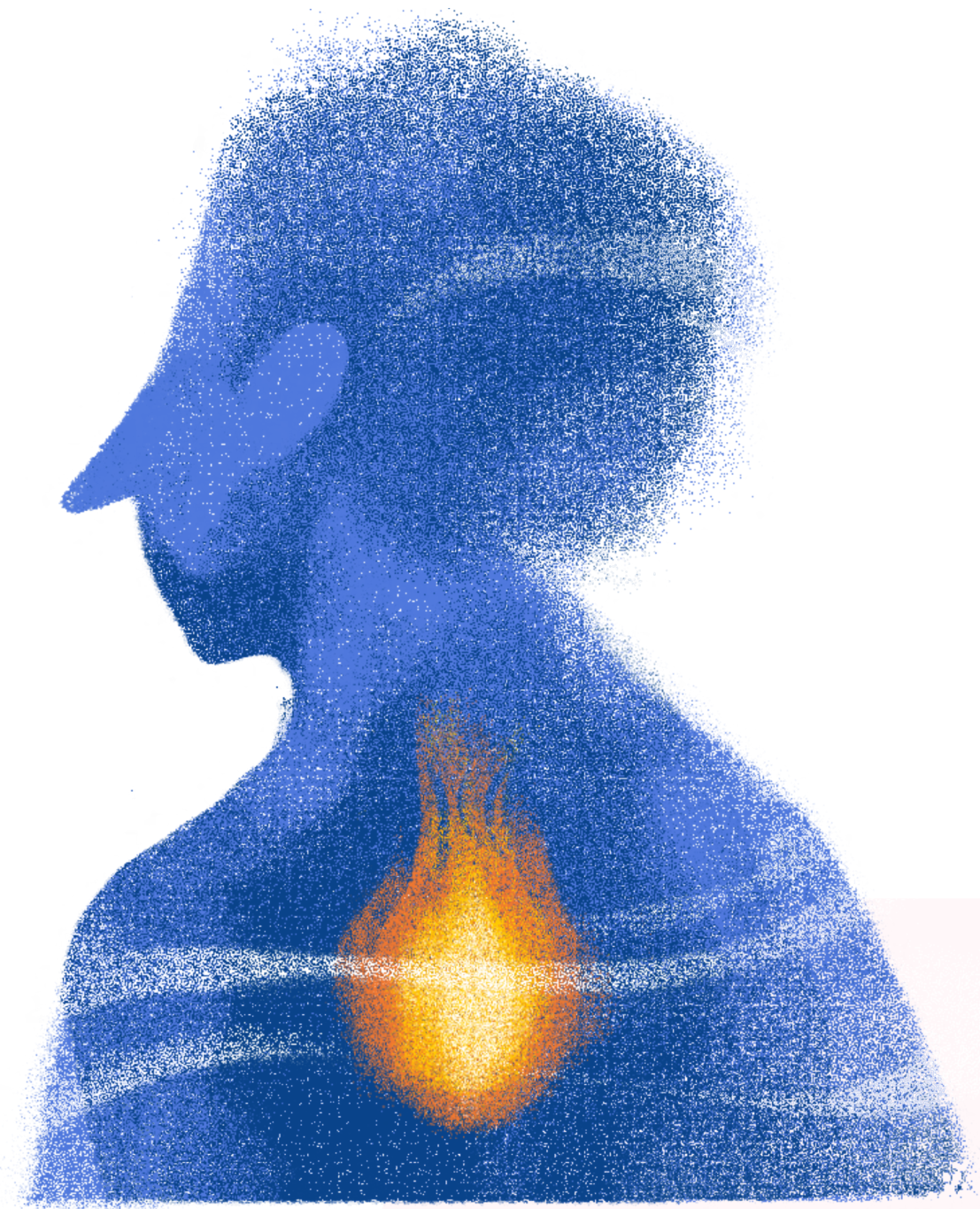
Discussion

What are the design implications
keeping attention in mind?



Discussion

What are the design implications
keeping attention in mind?



**Too many popups, Multiple competing
alerts, Simultaneous animations**



04 Cognitive Load

The total mental effort required to complete a task

Intrinsic Load

Extraneous Load

Task complexity itself
Learning recursion

04 Cognitive Load

The total mental effort required to complete a task

Intrinsic Load

Extraneous Load



Bad design adds difficulty
Confusing layout, Poor labeling

04 Cognitive Load

The total mental effort required to complete a task

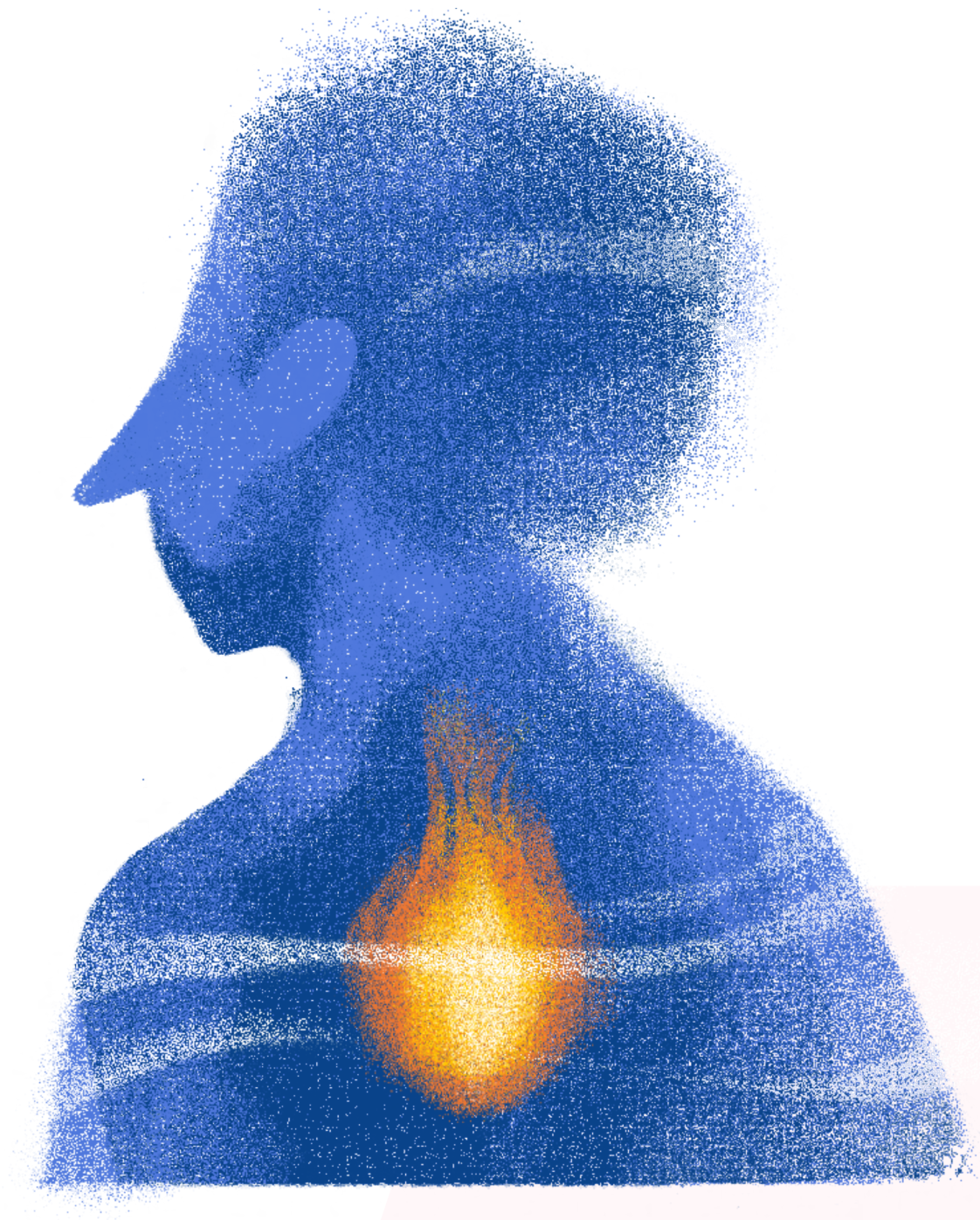
Intrinsic Load

Extraneous Load



Discussion

Why do checkout flows use multiple pages instead of one long form?



Mental Models vs System Models



Mental Models vs System Models

Mental Model: User's internal understanding of system

System Model: Actual internal logic of system

When They Don't Match → Errors

Example:

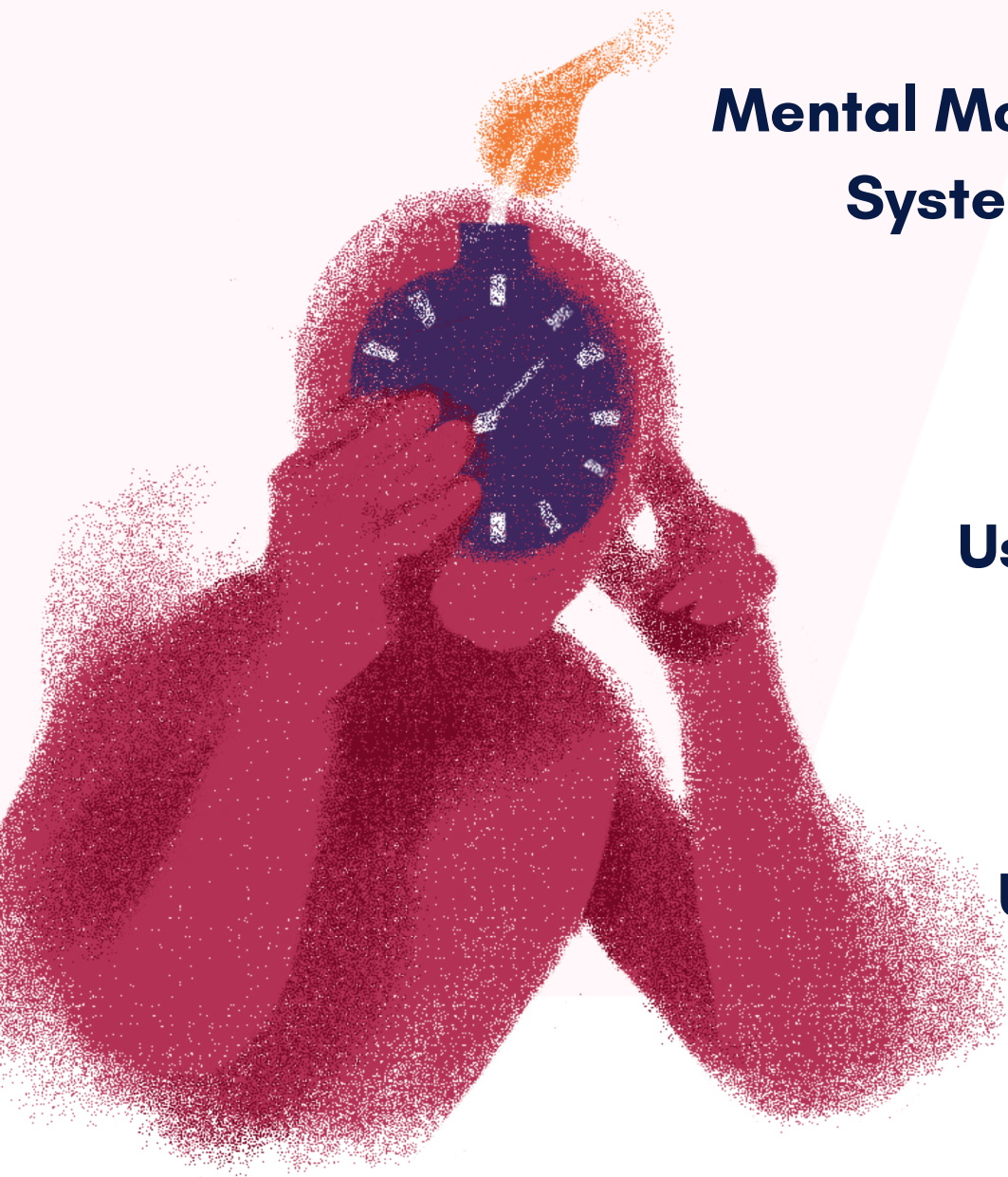
User thinks: Trash deletes permanently

But system: Moves to recycle bin

Example:

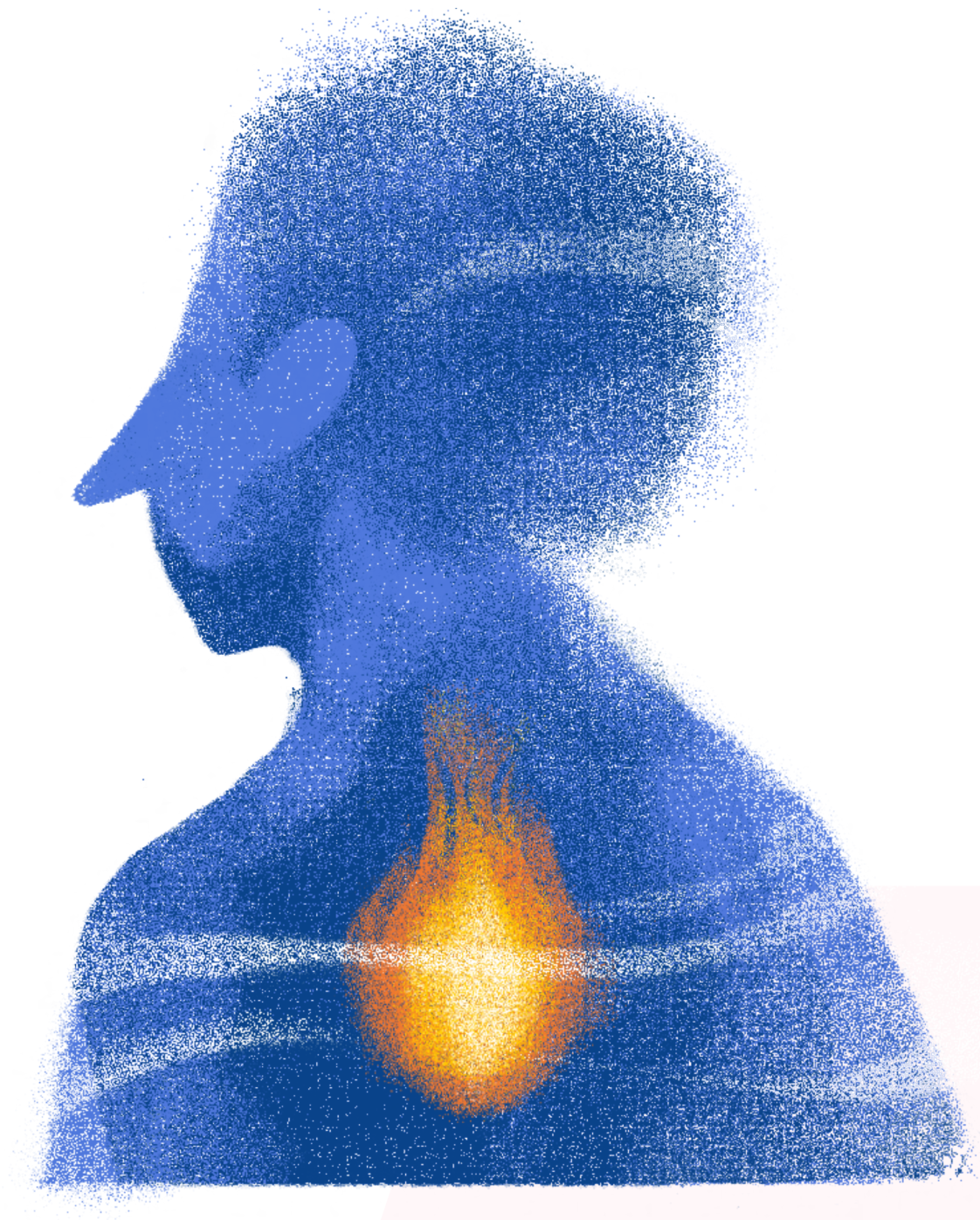
User drags file out of folder to copy

System moves it instead



Discussion

Why was the desktop metaphor so successful historically?



05 Error Patterns



Slips (Execution Errors)

Mistakes (Decision Errors)

Lapses (Memory Errors)

**User intends correct action but executes wrong
one due to attention failure
Clicking Delete instead of Save, Typo while typing
password**

05 Error Patterns



Slips (Execution Errors)

Mistakes (Decision Errors)

Lapses (Memory Errors)

Wrong plan chosen due to wrong mental model
Choosing wrong menu option, Misunderstanding icon meaning
Cause:

05 Error Patterns



Slips (Execution Errors)

Mistakes (Decision Errors)

Lapses (Memory Errors)

Forgetting step.
Forget to attach file, skip required field

05 Error Patterns



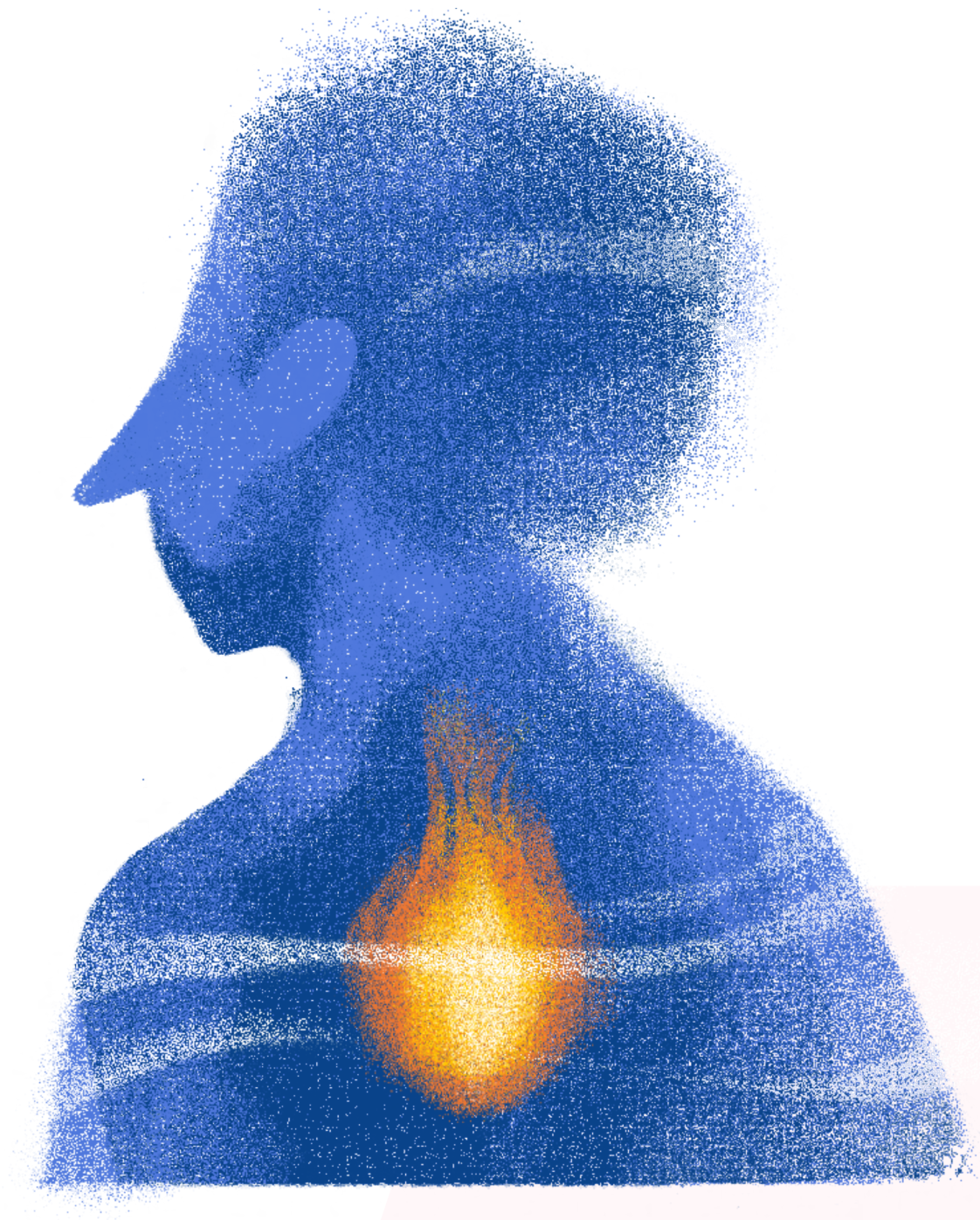
Slips (Execution Errors)

Mistakes (Decision Errors)

Lapses (Memory Errors)

Discussion

If many users keep making the same mistake, who is wrong?



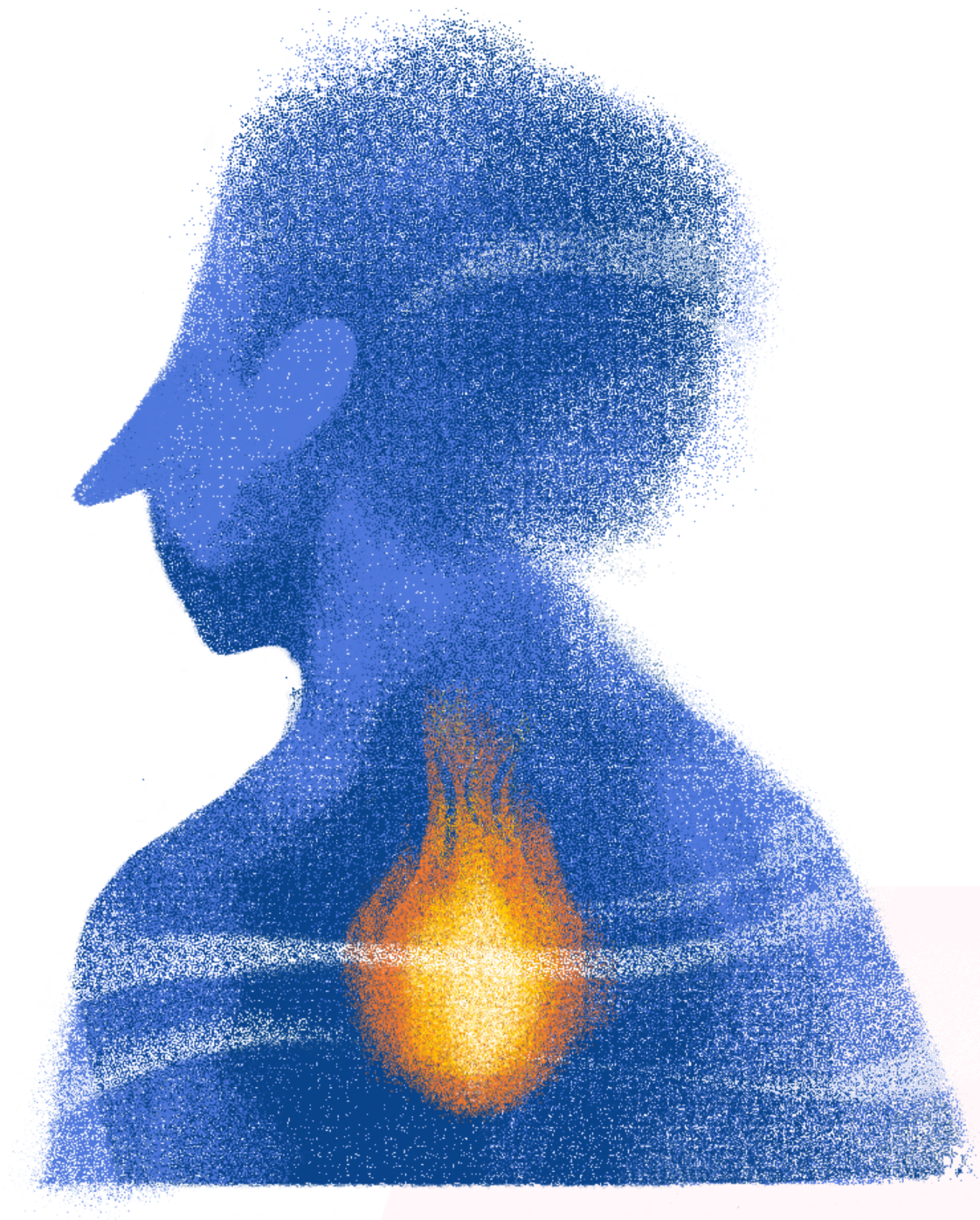
Discussion

How can we

Prevent slips

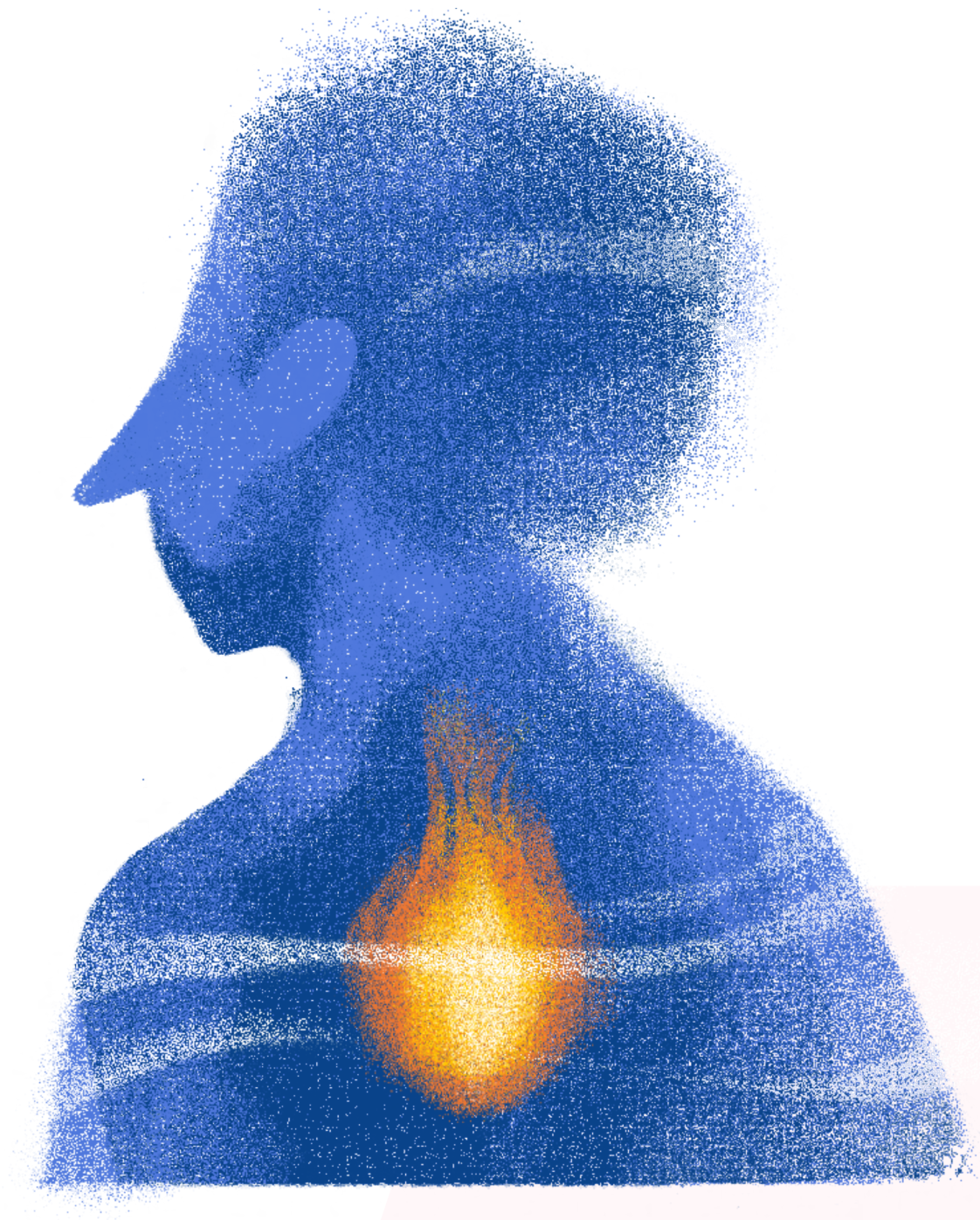
Prevent mistakes

Prevent lapses



Discussion

Why is “Undo” one of the most powerful UI features?



Every UI interaction follows a psychological process — whether designers realize it or not.

If we don't design for these steps, users get confused.

Norman's Action Cycle

Form goal

Form intention

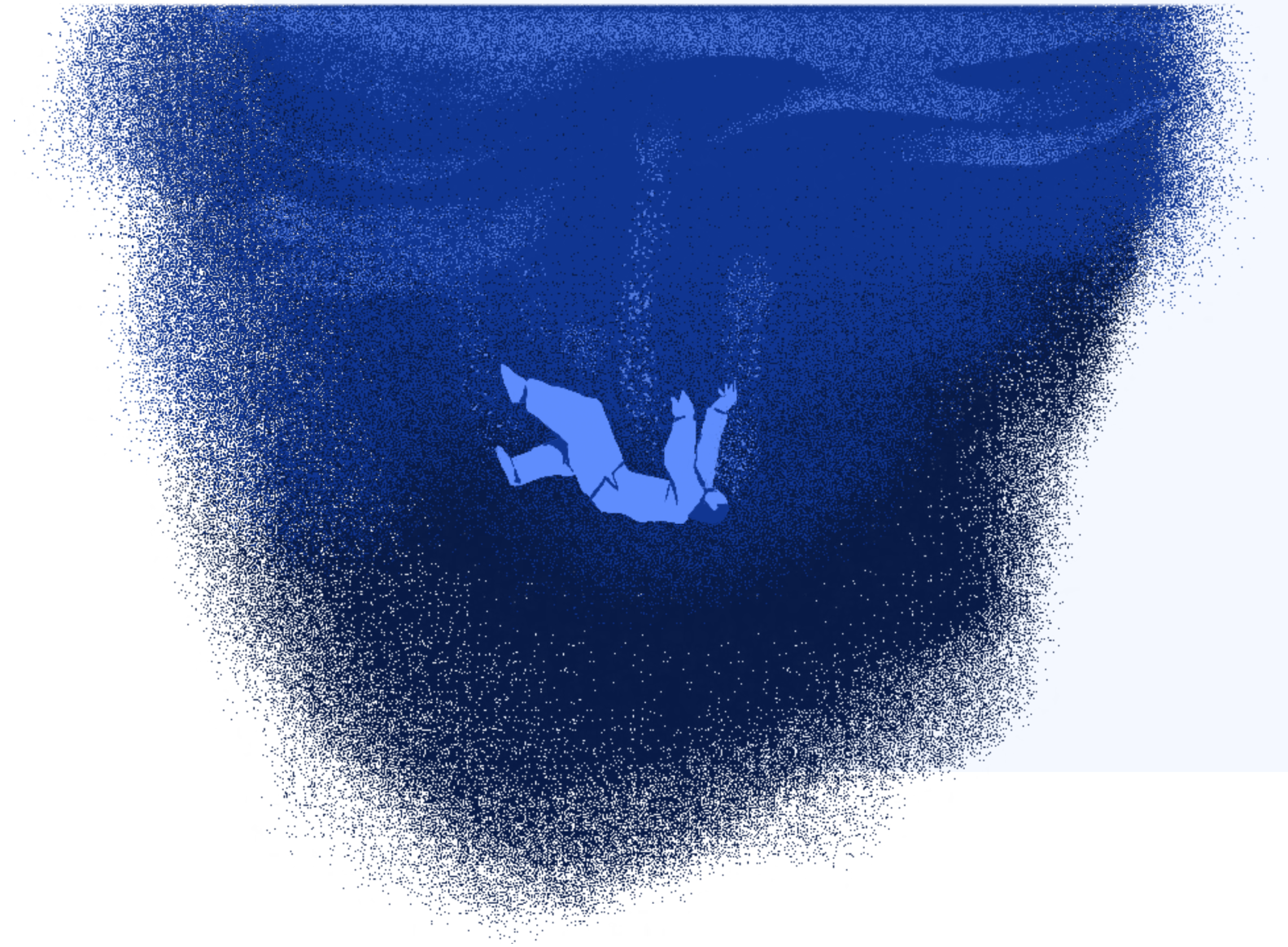
Specify action

Execute action

Perceive system state

Interpret system state

Evaluate outcome



Every usability problem happens at one of these steps.
Not randomly.
At a specific cognitive breakdown point.

Norman's Action Cycle

Form goal

Form intention

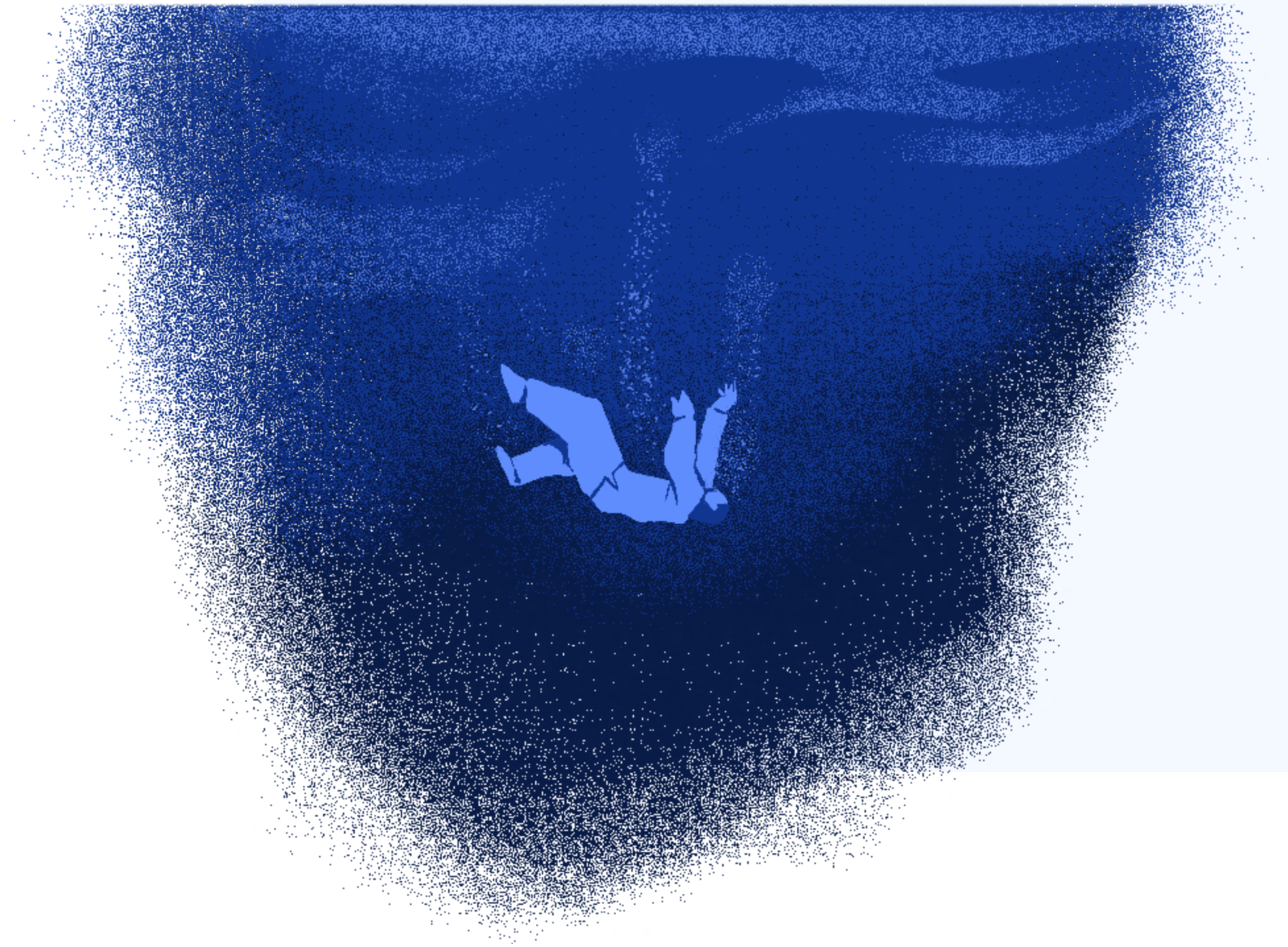
Specify action

Execute action

Perceive system state

Interpret system state

Evaluate outcome



This helps us:

- Diagnose usability problems systematically**
- Identify where cognition breaks**
- Justify design changes scientifically**
- Design intentionally instead of randomly**

Without it:

Based on aesthetics only

Based on personal opinion

Based on guesswork

Why users struggle?

Where breakdown occurs?

What to fix?

Norman's Model helps us identify where interaction breaks in the user's cognitive process.

Translate Ordering food online" to Norman's 7 stages

At what stage will they face the following problems:

Can't find restaurant

Payment error unclear

No confirmation message

Norman's Model helps us identify where interaction breaks in the user's cognitive process.

The “Where is the Button?” Problem

They cannot find the button

Can you explain the issue using the 7 steps?

“I Clicked It But Nothing Happened” Problem

They are not sure if an action took place

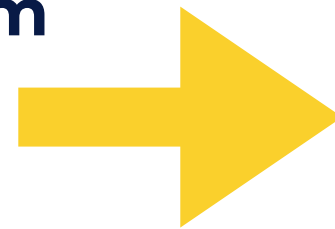
Can you explain the issue using the 7 steps?

Reduce both gulfs with clear controls and clear feedback

The “Where is the Button?” Problem

They cannot find the button

Can you explain the issue using the 7 steps?

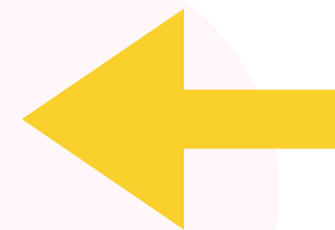


Gulf of Execution
Gap between what user wants
and
what system allows

“I Clicked It But Nothing Happened” Problem

They are not sure if an action took place

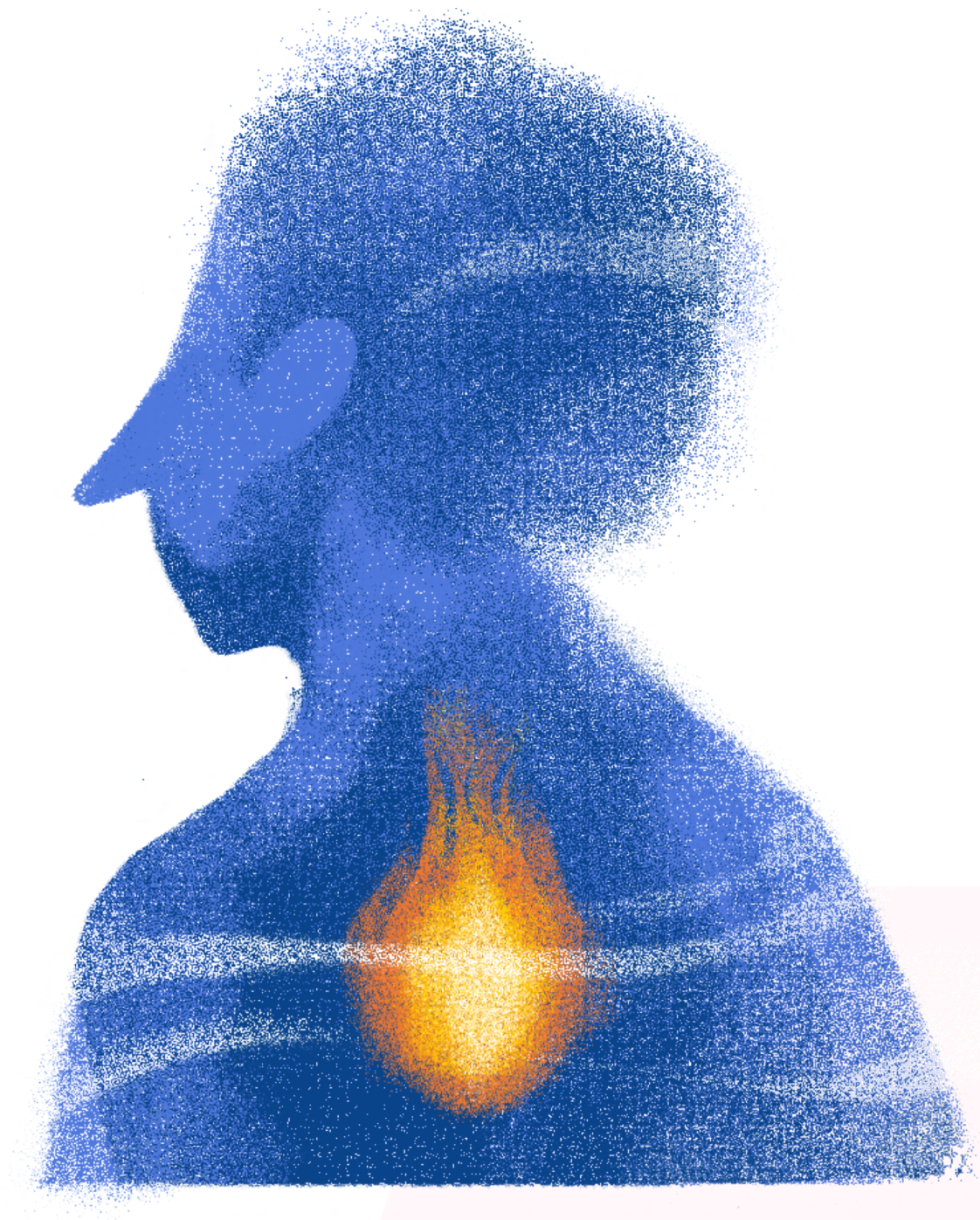
Can you explain the issue using the 7 steps?



Gulf of Evaluation
Gap between system output
and
user understanding

Discussion

**Which is worse: hard-to-find button
or unclear feedback?**



Choose one of the following apps and write the given deliverables:

ATM

Food delivery app

University LMS

Ride-hailing app

Deliverables:

user mental model diagram

system model diagram

mismatch points

redesign suggestion

Diagnose the issues using the following

perception

attention

memory

cognitive load

error patterns

Norman's gulfs

mental model mismatch

Accessibility Audit:

Visual

Contrast OK?

Text scalable?

Keyboard

Fully navigable?

Screen Reader

Labels present?

Forms

Errors clear?

Color Use

Color-only signals?

Motion

Excessive animation?

The end

